Selecting the photon vacuum

A coupling that distinguishes the points of the compact family of vacua, at the third rung where colour survives

Investigation note to *The electromagnetic sphere*

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Abstract

Contents

1 The compact families at the third rung **1**
1.1 The second rung, restated . . . 2
1.2 Four-generator involutions at the third rung . . . 3
1.3 The vacuum Π_A . . . 6
1.4 Kinetic term, current and transport on F_3 . . . 7
1.5 Fixed sets and the colour condition . . . 9
1.6 The answer . . . 10

2 Which charge is the photon, and the residual choice **10**
2.1 Two charges on the matter . . . 12
2.2 The counts against the normal $U(1)$, and the dichotomy . . . 13
2.3 Identification at the aligned vacua . . . 13
2.4 The residual choice . . . 16
2.5 Robustness . . . 18
2.6 The complete selection . . . 19
2.7 The answer . . . 20

1 The compact families at the third rung

The second-rung vacuum of [1, Def. 26.5] lies on a compact family S^4 of Lorentz-invariant projectors, and a spectrum-only potential is constant on it [1, Section 26]. A coupling that distinguishes its points has to be found at $n \geq 3$, where colour survives [1, Section 31]. This section fixes the geometry on which any such coupling acts. The results are as follows. At the third rung the Lorentz-invariant symmetric involutions that are products of four internal generators form three orbits of the compact group K. Only the four-positive orbit F_3, an oriented Grassmannian of dimension 16, commutes with colour, and it commutes with colour at every point (fixed). The second-rung family is one Clifford four-sphere inside F_3 and is not K-invariant (fixed). The vacuum Π_A built on the auxiliary four-plane is the analogue of [1, Def. 26.5]. It is neutral under the construction's charge and under electromagnetism, and it is fixed by the Pati–Salam and Standard-Model algebras (fixed). Its sixteen fluctuations are four complex Standard-Model doublets of hypercharge $\frac{1}{2}$ (fixed). Any K-invariant coupling is constant on F_3. The alignment energy of a proper gauged subalgebra vanishes exactly on its fixed set, and these fixed sets are determined here. A K-equivariant duality $P \mapsto P^{\perp}$ preserves every

gauge-boson mass matrix, so no alignment energy separates Π_A from Π_W (fixed). On the matter block the construction's charge acts as half the fermion number (fixed), and with $\mathfrak{su}(2)_L$ it is anomalous there (restated). Identifiers p1-n refer to the `PASS` lines of `verify/p1_families.py` (Table 1).

Conventions. The realisation is that of [1, App. A]: $E_{6(k-1)+a} = \omega^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}^{\otimes(n-k)}$ on $\mathbb{R}^{8^n}$, every generator and blade a signed permutation matrix, and $e_a^\top = \eta_{aa} e_a$. Products, commutators and traces of blades are therefore integer computations, and blades obey the commutation rule, equation (3) of [1], and the transpose rule, equation (4) there.

Definitions. Rung $n = 3$, spectator interior action: interior generators $\gamma^\mu = (E_1, E_4, E_5, E_6)$; internal generators the other fourteen. The positive internal span is $V_+ = \mathrm{span}(A \cup W)$, of dimension 8, and the negative internal span is $V_- = \mathrm{span}(C)$, of dimension 6. The groups are those of [1, Section 8]: auxiliary $A = (E_2, E_3, E_8, E_9)$, weak $W = (E_7, E_{13}, E_{14}, E_{15})$, written $(w_1, \ldots, w_4)$, and colour $C = (E_{10}, E_{11}, E_{12}, E_{16}, E_{17}, E_{18})$. $K = \mathrm{Spin}(8) \times \mathrm{Spin}(6)$ is the group generated by the 43 compact internal bivectors [1, Thm. 2.3], with Lie algebra $\mathfrak{k} = \mathfrak{so}(8) \oplus \mathfrak{so}(6)$. For an oriented four-plane P in V_+ with oriented orthonormal basis (u_1, u_2, u_3, u_4), $\mathrm{vol}_P := u_1u_2u_3u_4$, a symmetric involution commuting with every interior gamma, and $\Pi_P := \frac{1}{2}(\mathbf{1} + \mathrm{vol}_P)$. The compact family $F_3 := \{\Pi_P : P \text{ an oriented four-plane in } V_+\}$ is the K-orbit of Π_A (A the auxiliary four-plane) and is diffeomorphic to the oriented Grassmannian of four-planes in $\mathbb{R}^8$, of dimension 16 (Theorem 1.5). The gauged subalgebras considered are: $\mathfrak{h}_{43} = \mathfrak{k}$, all compact internal bivectors, as in [1, Prop. 21.6]; $\mathfrak{h}_{\mathrm{PS}} = \mathfrak{so}(4)_W \oplus \mathfrak{so}(6)_C$, the Pati–Salam algebra $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R \oplus \mathfrak{su}(4)$; $\mathfrak{h}_{\mathrm{SM}} = \mathfrak{su}(3)_C \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$, embedded as in [1, Section 8]; $\mathfrak{u}(1)_{\mathrm{em}}$, the bivector part of $Q = -JT_L^3 + Y$; and $\mathfrak{u}(1)_A = \mathrm{span}(q_A)$, the construction's charge $Q = \frac{1}{2}J_1$ at $n = 3$. The alignment energy of a gauged subalgebra $\mathfrak{h}$ with orthonormal basis (T_a) and couplings g_a is $V_{\mathfrak{h}}(\Pi) := \sum_a g_a^2 \|[T_a, \Pi]\|^2$ (Frobenius norm on $\mathbb{R}^{512}$). With the projector kinetic term $\frac{f^2}{2}\mathrm{Tr}(D\Pi D\Pi)$ of [1, Section 26] it is proportional to the trace of the gauge-boson mass matrix at Π.

Notation. J is the ordered product of the internal generators, $\Gamma = J\gamma^0\gamma^1\gamma^2\gamma^3$, $q_A = E_2E_3$, $Q_A = \frac{1}{2}q_A$, $\chi_W = w_1w_2w_3w_4$, χ_C the product of C, and $P_* = \frac{1}{8}(\mathbf{1} + X)(\mathbf{1} + Z)(\mathbf{1} + U)$ with $X = \Gamma\chi_W$, $Z = -J\chi_C$, $U = -Jq_A$ [1, Thm. 8.1]; $P_{L,R} = P_*\frac{1}{2}(\mathbf{1} \mp \Gamma)$. The colour Cartan element is $H_C = -(E_{10}E_{16} + E_{11}E_{17} + E_{12}E_{18})$, and $\mathfrak{su}(3)_C$ is its derived centraliser in $\mathfrak{so}(6)_C$ [1, Prop. 8.4]. Further $\tau_i = -\frac{1}{2}w_jw_k$ for cyclic (i, j, k), $t_i^{L,R} = \frac{1}{2}(\mathbf{1} \mp \chi_W)\tau_i$ [1, Thm. 8.2], the hypercharge bivector is $t_Y = t_3^R + \frac{1}{6}H_C$, and

$$B_{\mathrm{em}} = \tau_3 + \tfrac{1}{6}H_C = t_3^L + t_Y = -\tfrac{1}{2}E_7E_{13} + \tfrac{1}{6}H_C$$

spans $\mathfrak{u}(1)_{\mathrm{em}}$: on the block $Q = -J(t_L^3 + t_R^3 + \frac{1}{6}H_C)$ [1, Prop. 8.4]. The vector representation ρ is defined by $[X, u] = \rho(X)u$ for internal vectors u. Thus $\rho(\frac{1}{2}e_ie_j) = E_{ij} - E_{ji}$ on V_+, with a sign -1 on V_-, and the lift $M \mapsto \sum_{i<j} M_{ij}\frac{1}{2}e_ie_j$ (sign -1 on V_-) is a Lie algebra map inverse to ρ (p1-12). We write $\rho_+(X) \in \mathfrak{so}(8)$ for the V_+ part, p_P for the orthogonal projector of $\mathbb{R}^8$ onto P, θ for $+1$ on V_+ and -1 on V_-, and $\widetilde{\mathrm{Gr}}_k(V)$ for the oriented Grassmannian. On vectors, $\rho(Q_A)$ maps $E_2 \mapsto -E_3$, $E_3 \mapsto E_2$ and kills E_8, E_9, W. $\rho(B_{\mathrm{em}})$ maps $w_1 \mapsto w_2$, $w_2 \mapsto -w_1$ and kills A, w_3, w_4. $J_Y = 2\rho(t_Y)$ is a complex structure on W that kills A.

1.1 The second rung, restated

Proposition 1.1 (The second-rung family is a K_2-orbit; restated and fixed)**.** *At $n = 2$ the internal complement of the spectator action has signature $(5, 3)$, with $V_+^{(2)} = \mathrm{span}(E_2, E_3, E_7, E_8, E_9)$ and $V_-^{(2)} = \mathrm{span}(E_{10}, E_{11}, E_{12})$. Let $(a, b, c, d, e) = (E_2, E_3, E_7, E_8, E_9)$, $Q = \frac{1}{2}ab$, $H = abcd$, $\Pi_0 = \frac{1}{2}(\mathbf{1} + H)$, $I_5 = abcde$, $Z_i = I_5v_i$.*

(i) (Restated.) *[1, Thms. 26.6–26.8] hold: H is a symmetric traceless involution commuting with the interior gammas, Q and the internal volume; $[\frac{1}{2}be, \Pi_0] = -\frac{1}{2}acde$; the Z_i are a symmetric Clifford system, $T_i = [\frac{1}{2}v_i e, \Pi_0] = \frac{1}{2}Z_i$ with $\mathrm{Tr}(T_iT_j) = 16\delta_{ij}$, Q rotates (T_a, T_b) and fixes T_c, T_d; along $\Pi(n) = \frac{1}{2}(\mathbf{1} + I_5 n)$, $\mathrm{Tr}(d\Pi^2) = 16|dn|^2$, $-\mathrm{Tr}([Q,\Pi]d\Pi) = 16(n_a dn_b - n_b dn_a)$ and $[\Pi, d\Pi] = \frac{1}{4}[n, dn]$. The stabiliser of H in $\mathfrak{so}(5,3)$ is $\mathfrak{so}(P) \oplus \mathfrak{so}(P^\perp) = \mathfrak{so}(4) \oplus \mathfrak{so}(1,3)$, of dimension 12 with compact part 9, and every other internal bivector anticommutes with H [1, Prop. 26.9].*

(ii) (Fixed.) *The 13 compact internal bivectors preserve $\mathrm{span}\{Z_i\}$: $\mathfrak{so}(5)$ acts on it by the vector representation and $\mathfrak{so}(3)$ trivially. The K_2-orbit of Π_0, $K_2 = \mathrm{Spin}(5) \times \mathrm{Spin}(3)$, is the whole family $S^4 = \{\Pi(n)\}$, and $\exp(\frac{\vartheta}{2}v_i e)$ carries $\Pi(e)$ to $\Pi(\cos\vartheta\, e + \sin\vartheta\, v_i)$. Moreover $I_5 n = \mathrm{vol}_{n^\perp}$, so $S^4 = \{\Pi_P : P \in \widetilde{\mathrm{Gr}}_4(V_+^{(2)})\}$.*

(iii) (Fixed.) *$[Q, \Pi(n)] = 0$ exactly on the two-sphere $n_a = n_b = 0$, the planes containing $\mathrm{span}(a,b)$. Every $\Pi(n)$ has spectrum $\{0^{32}, 1^{32}\}$, and $\frac{1}{64}\sum \|[e_i e_j, \Pi(n)]\|^2 = 4$ over the 13 compact blades. A spectrum-only potential and the full alignment energy are constant on S^4.*

Proof. (i) is the computation of [1, Section 26], repeated in p1-1 to p1-4 and p1-7. The stabiliser statement is equation (3) of [1]: a bivector e_ie_j and the grade-four blade H commute iff $|\{i,j\} \cap \{a,b,c,d\}|$ is even. (ii) A positive bivector inside $V_+^{(2)}$ shares two indices with I_5 and commutes with it, so $[X, Z_i] = I_5[X, v_i] = I_5\rho(X)v_i$; a negative bivector shares none and commutes with each Z_i. Since $\mathrm{Spin}(5)$ is transitive on S^4, the orbit of $\Pi(e)$ is the family. The rotation formula is $\mathrm{Ad}(\exp\frac{\vartheta}{2}v_i e)e = \cos\vartheta\, e + \sin\vartheta\, v_i$. If $(f_1, \dots, f_4, n)$ is an oriented orthonormal basis of $V_+^{(2)}$, then $I_5 = f_1f_2f_3f_4n$ and $I_5 n = f_1f_2f_3f_4$ (p1-5). (iii) $[Q, Z_i] = I_5\rho(Q)v_i$, and $\rho(Q)$ rotates (a,b) and kills c, d, e (p1-6). Each $\Pi(n)$ is conjugate to Π_0 by an orthogonal matrix, and the alignment sum is K_2-invariant, as in Proposition 1.15 below (p1-4, p1-5). □

At $n = 2$ three descriptions coincide: the Clifford sphere of five positive generators, the K_2-orbit, and the oriented Grassmannian of four-planes in $V_+^{(2)}$. At $n = 3$ they separate. The Grassmannian description is the one that continues.

1.2 Four-generator involutions at the third rung

Lemma 1.2 (Blade census; fixed)**.** *At $n = 3$ the eighteen generators satisfy the Clifford relations of $\mathbb{R}^{9,9}$, $V_+ = \mathrm{span}(E_2, E_3, E_7, E_8, E_9, E_{13}, E_{14}, E_{15})$, and the 43 compact internal bivectors square to $-\mathbf{1}$ while the 48 mixed ones square to $+\mathbf{1}$ (p1-8).*

(i) Of the 3060 blades of grade four exactly the 1001 internal ones commute with the four interior gammas. All are traceless. One with q negative factors is a symmetric involution if q is even and an antisymmetric complex structure if q is odd. The counts are $(4,0) : 70$, $(3,1) : 336$, $(2,2) : 420$, $(1,3) : 160$, $(0,4) : 15$ (p1-9).

(ii) More generally an internal blade of even grade r with q negative factors is a symmetric involution iff $r(r-1)/2 + q$ is even. The only nontrivial one invariant under K is $\mathrm{vol}_+ = E_2E_3E_7E_8E_9E_{13}E_{14}E_{15}$, and $\mathrm{vol}_+ = -Z = J\chi_C$ (p1-10).

Proof. For $|I| = 4$, equation (3) of [1] gives $e_I e_j = (-1)^{4-|I\cap\{j\}|}e_j e_I$, so a generator commutes with e_I iff it is not a factor. By equation (4) of [1], $e_I^2 = (-1)^{r(r-1)/2+q}\mathbf{1}$ and $e_I^\top = e_I^2 e_I$. An even internal blade commutes with e_ie_j iff $|I \cap \{i,j\}|$ is even. For all same-sign pairs this forces $I \cap V_\pm \in \{\emptyset, V_\pm\}$, which leaves vol_+, χ_C ($r = q = 6$, square $-\mathbf{1}$) and $\pm J$ ($r = 14$, square $-\mathbf{1}$). The identity $\mathrm{vol}_+ = J\chi_C$ is checked on the matrices. □

Lemma 1.3 (Transpose and recovery; fixed)**.** *Let P be a nondegenerate internal four-plane with η-orthonormal oriented basis $(u_1, \dots, u_4)$ and $\mathrm{vol}_P = u_1u_2u_3u_4$. Then vol_P commutes with the interior gammas, $\mathrm{vol}_P^2 = (-1)^q\mathbf{1}$, the internal vectors anticommuting with vol_P are exactly those of P, and*

$$\mathrm{vol}_P^\top = \mathrm{vol}_{\theta P}.$$

Hence for q even, Π_P is an orthogonal projector iff $\theta P = P$, that is iff $P = (P \cap V_+) \oplus (P \cap V_-)$. For a positive-definite $P \not\subset V_+$, Π_P is an idempotent but not symmetric (p1-13).

Proof. The interior gammas anticommute with every internal vector, hence commute with a product of four. By equation (4) of [1], $\mathrm{vol}_P^2 = (-1)^6 \prod \eta(u_i)\mathbf{1} = (-1)^q\mathbf{1}$. For an internal vector u, $u^\top = \theta u$. Transposing reverses the product, and reversing four anticommuting vectors costs $(-1)^6 = 1$. So $\mathrm{vol}_P^\top = \theta u_1 \cdots \theta u_4 = \mathrm{vol}_{\theta P}$, and θ is an isometry of η. Each u_i commutes with itself and anticommutes with the other three factors, so every vector of P anticommutes with vol_P. A vector η-orthogonal to P anticommutes with all four factors, so it commutes with vol_P. Since P is nondegenerate, an internal v splits as $v_P + v_\perp$; if v anticommutes with vol_P, so does $v_\perp$, which also commutes, hence $v_\perp \mathrm{vol}_P = 0$ and $v_\perp = 0$. So P is determined by vol_P, and its orientation by the sign. Then vol_P is symmetric iff $\theta P = P$ as oriented planes. □

Lemma 1.4 (Colour acts irreducibly on V_-; fixed)**.** *The centraliser of $\rho(H_C)$ in $\mathfrak{so}(V_-)$ is $\mathfrak{u}(3)$. Its part orthogonal to H_C is $\mathfrak{su}(3)_C$, of dimension 8 and perfect. Lifted to $\mathfrak{k}$, $\mathfrak{su}(3)_C$ commutes with H_C and acts trivially on V_+. Its commutant on $V_- = \mathbb{R}^6$ is $\mathrm{span}(\mathbf{1}, \frac{1}{2}\rho(H_C))$ with $(\frac{1}{2}\rho(H_C))^2 = -\mathbf{1}$, a field, so $\mathfrak{su}(3)_C$ acts irreducibly on V_- (p1-15).*

Proof. The computation is in the six-dimensional vector model and is lifted by the Lie algebra map of p1-12. A proper invariant subspace of an orthogonal representation has an invariant orthogonal complement, and the projector onto it is an idempotent in the commutant other than 0 and $\mathbf{1}$. A field has no such idempotent. □

Theorem 1.5 (The three orbit types and the colour test; fixed)**.** *The Lorentz-invariant symmetric involutions vol_P, with P a nondegenerate internal four-plane, form three K-orbits. Their stabilisers and dimensions are:*

type	*orbit*	*stabiliser in $\mathfrak{so}(8,6)$*	*dim*	*compact part*	*orbit dim*
$(4,0)$	$F_3 \cong \widetilde{\mathrm{Gr}}_4(V_+)$	$\mathfrak{so}(4) \oplus \mathfrak{so}(4,6)$	51	27: $\mathfrak{so}(4)_A \oplus \mathfrak{so}(4)_W \oplus \mathfrak{so}(6)_C$	16
$(2,2)$	$\widetilde{\mathrm{Gr}}_2(V_+) \times \widetilde{\mathrm{Gr}}_2(V_-)/\pm$	$\mathfrak{so}(2,2) \oplus \mathfrak{so}(6,4)$	51	23	20
$(0,4)$	$\widetilde{\mathrm{Gr}}_4(V_-)$	$\mathfrak{so}(4) \oplus \mathfrak{so}(8,2)$	51	35	8

Every point has spectrum $\{+1^{256}, -1^{256}\}$, so no spectrum-only potential separates the points or the types. Every point of F_3 commutes with all of $\mathfrak{so}(6)_C \supset \mathfrak{su}(3)_C$. No point of the $(2,2)$ or $(0,4)$ orbits commutes with $\mathfrak{su}(3)_C$.

Proof. By Lemma 1.3 the symmetric ones are the $\mathrm{vol}_{P_+}\mathrm{vol}_{P_-}$ with $P_\pm \subset V_\pm$ of dimensions (p,q), q even. For $g \in K$, $\mathrm{Ad}(g)u = \rho(g)u$ on vectors, so $\mathrm{Ad}(g)\mathrm{vol}_P = \mathrm{vol}_{\rho(g)P}$ (p1-14). $SO(8) \times SO(6)$ is transitive on pairs of oriented planes, which gives one orbit per type; reversing both orientations in type $(2,2)$ gives the same element. By the recovery statement of Lemma 1.3, $\exp(tX)$ fixes vol_P iff $\rho(X)$ preserves P, so the stabiliser is $\mathfrak{so}(P) \oplus \mathfrak{so}(P^\perp)$. Bivector by bivector, e_ie_j commutes with vol_P iff both or neither of i, j lie in P, and anticommutes otherwise; the counts are p1-11. The orbit dimension is 43 minus the compact part. A symmetric involution of trace zero has eigenvalues ± 1, each 256 times (p1-42). A vector of V_+ anticommutes with both factors of a colour bivector and so commutes with it; hence every colour bivector commutes with every vol_P, $P \subset V_+$. For types $(2,2)$ and $(0,4)$, $\mathfrak{su}(3)_C$ fixes vol_P iff it preserves the plane $P_- \subset V_-$ of dimension 2 or 4, which Lemma 1.4 excludes. p1-16 checks this at the representatives and at random K-translates. □

At $n = 3$ the stabiliser of the four-positive type is the row of [1, Prop. 26.9]. The same proof covers all even grades: a θ-stable plane of type (p, q) with $r(r-1)/2 + q$ even gives a K-orbit of dimension $p(8-p) + q(6-q)$. Among them only $\mathrm{vol}_+ = -Z$ is K-fixed (Lemma 1.2). No compact generator moves $\frac{1}{2}(\mathbf{1} + \mathrm{vol}_+)$, as for the chirality projectors of [1, Prop. 26.4]; only the boosts do. Lemma 1.3 agrees with [1, Thm. 11.3], by which a variation $[B, \Pi]$ stays among the orthogonal projectors exactly when B is compact: a noncompact generator tilts P out of V_+, and Π_P stops being symmetric.

Proposition 1.6 (Clifford spheres in F_3; fixed)**.** *(i) For an oriented five-plane $L \subset V_+$ with oriented orthonormal basis $(v_1, \dots, v_5)$ put $I_L = v_1 \cdots v_5$ and $Z_i = I_L v_i$. The Z_i are a symmetric Clifford system, and $\frac{1}{2}(\mathbf{1} + I_L n) = \Pi_{n^\perp \cap L}$. Hence $S_L := \{\frac{1}{2}(\mathbf{1} + I_L n) : |n| = 1\} = \{\Pi_P : P \subset L\} \cong S^4$ is the $\mathrm{Spin}(L)$-orbit of any of its points. Dually, for an oriented three-plane U with volume U_3, $S^U := \{\Pi_P : P \supset U\} = \{\frac{1}{2}(\mathbf{1} + nU_3) : n \in U^\perp, |n| = 1\} \cong S^4$.*

(ii) Through each Π_P pass the spheres S_L with $L = P \oplus \mathbb{R}u$, $u \in P^\perp$ (a projective space $\mathbb{R}P^3$ of them), with tangent $P \otimes u$, and the spheres S^U with $U = P \cap a^\perp$, $a \in P$, with tangent $a \otimes P^\perp$. These are the two kinds of maximal linear spaces of rank-one elements in $T_{\Pi_P} F_3 \cong P \otimes P^\perp$. The inner spheres are parametrised by $\mathrm{Gr}_5(\mathbb{R}^8)$, of dimension 15, and none of them is K-invariant.

(iii) On the inner sphere through $A \oplus \mathbb{R}E_7$ with $Q = Q_A$, and on the outer sphere of planes containing $\mathrm{span}(E_3, E_8, E_9)$ with $Q = B_{\mathrm{em}}$, $\mathrm{Tr}(d\Pi^2) = 128|dn|^2$ and $-\mathrm{Tr}([Q, \Pi]d\Pi) = \pm 128(n_a dn_b - n_b dn_a)$. On the inner sphere $[\Pi, d\Pi] = \frac{1}{4}[n, dn]$. This is the structure of [1, Thms. 26.7, 26.8] with 16 replaced by 128 (p1-19, p1-32).

(iv) The second-rung family embedded with its own generators, $L_0 = \mathrm{span}(a, b, c, d, e) = \mathrm{span}(E_2, E_3, E_7, E_8, E_9)$, is the inner sphere S_{L_0}. Its point $n = c$ is Π_A, since $I_5 c = E_2E_3E_8E_9$, and its point $n = e$ is $\Pi_0 = \frac{1}{2}(\mathbf{1} + E_2E_3E_7E_8)$. On S_{L_0}, $Q_A = \frac{1}{2}ab$ is unbroken on the two-sphere $n_a = n_b = 0$. The algebras $\mathfrak{u}(1)_{\mathrm{em}}$, $\mathfrak{h}_{\mathrm{SM}}$ and $\mathfrak{h}_{\mathrm{PS}}$ are each unbroken only at $n = \pm c$, that is at $\Pi_{\pm A}$ (p1-47).

Proof. (i) As in Proposition 1.1(ii): $Z_iZ_j + Z_jZ_i = 2\delta_{ij}$ because I_L commutes with each v_i and squares to $+\mathbf{1}$, and $I_L n = f_1f_2f_3f_4$ for an oriented basis $(f_1, \dots, f_4, n)$ of L. For S^U, nU_3 is the volume of $\mathrm{span}(n, U)$; vectors of $U^\perp$ anticommute with U_3, so the nU_3 again form a Clifford system (p1-19). (ii) The orbit of $\mathrm{Spin}(L)$, respectively $\mathrm{Spin}(U^\perp)$, has tangent $[\mathfrak{m}_P \cap \mathfrak{so}(L), \Pi_P]$, respectively $[\mathfrak{m}_P \cap \mathfrak{so}(U^\perp), \Pi_P]$, where $\mathfrak{m}_P = P \wedge P^\perp$ is spanned by the bivectors mixing P and $P^\perp$. These intersections are $P \wedge u$ and $a \wedge P^\perp$. (iii) Π and $d\Pi$ are combinations of the Z_i, and $\mathrm{Tr}(Z_iZ_j) = 512\delta_{ij}$; $[Q, Z_i] = \pm Z_j$ by the action of $\rho(Q)$ on L or on $U^\perp$ (p1-32). (iv) The identities are blade identities. The fixed points follow from Theorem 1.16 below, since $P \subset L_0$ contains or is orthogonal to $\mathrm{span}(E_7, E_{13})$ only for $P = A$, and $E_{13} \notin L_0$; p1-47 computes the kernels of $n \mapsto [X, I_5 n]$ directly. □

Proposition 1.7 (The duality $P \mapsto P^\perp$; fixed)**.** vol_+ *commutes with $\mathfrak{k}$, with the interior gammas and with every vol_P, and $\mathrm{vol}_+\mathrm{vol}_A = \mathrm{vol}_W$. The map $D(\Pi) = \frac{1}{2}(\mathbf{1} + \mathrm{vol}_+(2\Pi - \mathbf{1}))$ sends Π_P to $\Pi_{P^\perp}$, with $P^\perp \subset V_+$ suitably oriented. It is a K-equivariant involution of F_3 without fixed points, and $D(\Pi_A) = \Pi_W$. For every $T, T' \in \mathfrak{k}$,*

$$\mathrm{Tr}([T, D\Pi][T', D\Pi]) = \mathrm{Tr}([T, \Pi][T', \Pi]), \qquad \mathrm{Tr}([T, \mathbf{1} - \Pi][T', \mathbf{1} - \Pi]) = \mathrm{Tr}([T, \Pi][T', \Pi]).$$

So every gauge-boson mass matrix, and every alignment energy $V_{\mathfrak{h}}$ with $\mathfrak{h} \subset \mathfrak{k}$, takes equal values at $\Pi_{\pm A}$ and $\Pi_{\pm W}$. On the block, $\mathrm{vol}_+ P_* = -P_*$, and $\mathrm{Tr}(\Pi_A P_L) = 32$, $\mathrm{Tr}(\Pi_W P_L) = 0$, $\mathrm{Tr}(\Pi_W P_R) = 32$ (p1-20, p1-46).

Proof. vol_+ is K-invariant by Lemma 1.2, and it commutes with the interior gammas, with which it shares no index. In an orthonormal basis $(u_1, \dots, u_8)$ of V_+ adapted to $P \oplus P^\perp$, $\mathrm{vol}_+ =$

$\pm u_1 \cdots u_8$, so vol_+ commutes with $\mathrm{vol}_P = u_1u_2u_3u_4$ and $\mathrm{vol}_+\mathrm{vol}_P = \pm u_5u_6u_7u_8 = \pm\mathrm{vol}_{P^\perp}$. D is an involution because $\mathrm{vol}_+^2 = \mathbf{1}$, and it has no fixed point because $P \neq P^\perp$. Since vol_+ commutes with T and Π, $[T, D\Pi] = \mathrm{vol}_+[T, \Pi]$, and $\mathrm{vol}_+^2 = \mathbf{1}$ gives the identity. The second identity is $[T, \mathbf{1} - \Pi] = -[T, \Pi]$. On the block, $\mathrm{vol}_+ P_* = -ZP_* = -P_*$. The traces follow from Proposition 1.11. □

1.3 The vacuum Π_A

Definition 1.8 (The third-rung vacuum)**.** $\Pi_A := \frac{1}{2}(\mathbf{1} + E_2E_3E_8E_9)$, built from the surplus pairs of the first and second constituents, and $\Pi_W := \frac{1}{2}(\mathbf{1} + \chi_W)$. Their orientation reversals are $\Pi_{-A} = \mathbf{1} - \Pi_A$ and $\Pi_{-W} = \mathbf{1} - \Pi_W$.

Theorem 1.9 (Lorentz invariant, neutral and not rigid; fixed)**.** *Π_A and Π_W are rank-256 orthogonal projectors with the following properties.*

(i) Each commutes with the interior gammas, with J, with q_A, with B_{em}, with the 21 generators of $\mathfrak{h}_{\mathrm{PS}}$ and with the 12 of $\mathfrak{h}_{\mathrm{SM}}$ (p1-21).

(ii) The compact internal bivectors that move Π_A are exactly the 16 that mix A with W. The tangents $T_{aw} = [\frac{1}{2}aw, \Pi_A] = \frac{1}{2}aw\,\mathrm{vol}_A$ are symmetric with $\mathrm{Tr}(T_{aw}T_{a'w'}) = 128\,\delta_{aa'}\delta_{ww'}$; the same holds at Π_W (p1-23).

(iii) The stabiliser of Π_A in $\mathfrak{k}$ is $\mathfrak{so}(4)_A \oplus \mathfrak{so}(4)_W \oplus \mathfrak{so}(6)_C$, of dimension 27. It contains $\mathfrak{u}(1)_A$, $\mathfrak{u}(1)_{\mathrm{em}}$, $\mathfrak{h}_{\mathrm{PS}}$ and $\mathfrak{h}_{\mathrm{SM}}$ (p1-35).

The literal embedding $\frac{1}{2}(\mathbf{1} + E_2E_3E_7E_8)$ of the second-rung vacuum is also a point of F_3 and commutes with q_A. It does not commute with B_{em} or with $\mathfrak{su}(2)_L$, because $E_7 \in W$ (p1-22).

Proof. All statements are equation (3) of [1] applied to blades. vol_A shares two indices with q_A, none with E_7E_{13}, the colour bivectors or the weak bivectors, and four with J. A bivector aw shares one index with vol_A and anticommutes with it, so $[\frac{1}{2}aw, \Pi_A] = \frac{1}{4}[aw, \mathrm{vol}_A] = \frac{1}{2}aw\,\mathrm{vol}_A$. Symmetry and the trace follow from $(aw\,\mathrm{vol}_A)^2 = -(aw)^2 = \mathbf{1}$ and the tracelessness of nontrivial blades. B_{em}, $t_i^{L,R}$ and $\mathfrak{su}(3)_C$ are combinations of bivectors inside W or C. □

In [1, Prop. 10.12] the compact spectator algebra carries colour on three negative planes, weak isospin on two positive planes and a family $\mathfrak{so}(4)$ on the remaining two. With the grouping of [1, Section 8] the remaining two planes are those of A, so Π_A separates the weak four-plane from the family four-plane and keeps everything else (interpretive). By Proposition 1.6(iv), Π_A is the point $n = c$ of the second-rung family itself, embedded at $n = 3$. The point $n = e$ chosen in [1, Def. 26.5] is not neutral at $n = 3$.

Theorem 1.10 (The fluctuations at Π_A; fixed)**.** *Let $T_{\Pi_A}F_3 = \mathrm{span}\{T_{aw}\} \cong A \otimes W$.*

(i) Under $\mathfrak{so}(4)_A \oplus \mathfrak{so}(4)_W \cong \mathfrak{su}(2)^{\oplus 4}$ it is the real irreducible $(\mathbf{2}, \mathbf{2}; \mathbf{2}, \mathbf{2})$, with commutant $\mathbb{R}$ and every $\mathfrak{su}(2)$ Casimir equal to $\frac{3}{4}$ (p1-24).

(ii) Under $\mathfrak{h}_{\mathrm{SM}}$, $\mathfrak{su}(3)_C$ acts trivially; $\mathfrak{su}(2)_L$ has Casimir $\frac{3}{4}$, and $\mathrm{ad}\,t_Y$ satisfies $(\mathrm{ad}\,t_Y)^2 = -\frac{1}{4}$ and commutes with $\mathfrak{su}(2)_L$. The space is four complex doublets $(\mathbf{1}, \mathbf{2})_{1/2}$, one for each direction of A, and $\mathfrak{so}(4)_A$ rotates the doublet index (p1-25).

(iii) Under $\mathfrak{u}(1)_{\mathrm{em}}$, $\mathrm{ad}\,B_{\mathrm{em}} = \mathrm{ad}(t_L^3 + t_Y)$. The charged directions are $A \otimes \mathrm{span}(w_1, w_2)$, four complex scalars $\phi_a = x_{a,13} + ix_{a,7}$ of charge one, the upper components of the doublets. The neutral directions are $A \otimes \mathrm{span}(w_3, w_4)$, eight real, the lower components.

(iv) Under $\mathfrak{u}(1)_A$ the charged directions are $\mathrm{span}(E_2, E_3) \otimes W$, four complex scalars $\phi_w = x_{2,w} + ix_{3,w}$. The neutral directions are $\mathrm{span}(E_8, E_9) \otimes W$, eight real.

(v) The same holds at Π_W *with* $\mathrm{Hom}(W,A)\cong W\otimes A$ *(p1-24, p1-25, p1-26).*

Here $x_{a,w}$ *are the coordinates of* $\Pi=\mathrm{Ad}(e^X)\Pi_A$, $X=\sum x_{a,w}\frac{1}{2}aw$. *At* $n=2$ *the same count gives one charged complex scalar and two neutral real scalars [1, Thm. 26.7].*

Proof. For X in the stabiliser, $[X,[Y,\Pi_A]]=[[X,Y],\Pi_A]$. So $T_{\Pi_A}F_3$ is the representation of the stabiliser on $\mathfrak{m}_A=A\wedge W\cong A\otimes W$, on which $\rho(\frac{1}{2}aw)$ is the rotation of the (a,w) plane. All five statements reduce to the action on A and W listed in the conventions. $\mathfrak{su}(2)_L$ acts on W with a quaternionic commutant (p1-37), so W is one complex doublet with complex structure $J_Y=2\rho(t_Y)$. $\rho(t_L^3+t_Y)=\rho(\tau_3)$ rotates (w_1,w_2) at unit rate and kills (w_3,w_4). p1-26 gives the explicit pairs, for example $[Q_A,T_{2,w}]=-T_{3,w}$, $[Q_A,T_{3,w}]=T_{2,w}$ and $[B_{\mathrm{em}},T_{a,7}]=T_{a,13}$, $[B_{\mathrm{em}},T_{a,13}]=-T_{a,7}$. □

Proposition 1.11 (The vacuum on the matter block; fixed, and restated)**.** *(i)* vol_A *and* χ_W *commute with* X,Z,U, *and* $\mathrm{vol}_A P_*=-\Gamma P_*$, $\chi_W P_*=\Gamma P_*$. *Hence* $\Pi_A P_*=P_L$ *and* $\Pi_W P_*=P_R$, *the two Weyl blocks of rank* 32 *(p1-27).*

(ii) $q_A P_*=JP_*$. *So* $Q_A=\frac{1}{2}q_A$, *the third-rung form of the construction's charge* $\frac{1}{2}J_1$, *acts on the block as* $\frac{1}{2}J$: *its charge operator* $-JQ_A$ *equals* $\frac{1}{2}$ *on all sixteen Weyl components. It is half the fermion number of [1, Section 8], not the electric charge, which takes the values* $-1,-\frac{1}{3},0,\frac{2}{3}$ *(p1-18, p1-27).*

(iii) Every T_{aw} *anticommutes with* $X=\Gamma\chi_W$, *and* $P_*T_{aw}P_*=0$: *the fluctuations of* Π_A *map the block into its orthogonal complement (p1-27).*

(iv) (Restated, [1, Prop. 8.5] with $a=\frac{1}{2}$, $b=0$.) *With* $\hat{Q}_A=-JQ_A$ *and* $\hat{T}=-Jt_3^L$, *we have* $\mathrm{Tr}(P_L\hat{Q}_A\hat{T}^2)=4$ *and* $\mathrm{Tr}(P_R\hat{Q}_A\hat{T}^2)=0$, *and likewise* 4 *for* $\mathfrak{su}(2)_R$ *on* P_R. *For* $\hat{Q}_{\mathrm{em}}=-JB_{\mathrm{em}}$, $\mathrm{Tr}(P_L\hat{Q}_{\mathrm{em}}\hat{T}^2)=0$. *The mixed* $SU(2)_L^2U(1)_A$ *anomaly does not cancel on the block, while the gravitational one does (p1-28) [8, 9].*

Proof. (i) $\mathrm{vol}_A=\mathrm{vol}_+\chi_W$ by Proposition 1.7. vol_+ commutes with Γ and $\mathrm{vol}_+P_*=-P_*$; with $\chi_W P_*=\Gamma P_*$ [1, Thm. 8.1] this gives $\mathrm{vol}_A P_*=-\Gamma P_*$. (ii) $UP_*=P_*$ means $-Jq_AP_*=P_*$, and $J^2=-\mathbf{1}$ gives $q_AP_*=JP_*$; then $-JQ_AP_*=\frac{1}{2}P_*$. (iii) T_{aw} is half a blade of grade four with three factors in A and one in W. It shares one index with χ_W and anticommutes with it, and it commutes with Γ, which is $\pm$ the product of all eighteen generators. Hence it anticommutes with X (p1-27). Since $XP_*=P_*$, $P_*TP_*=P_*\frac{1}{2}(\mathbf{1}+X)TP_*=P_*T\frac{1}{2}(\mathbf{1}-X)P_*=0$. (iv) On P_L, $\hat{Q}_A=\frac{1}{2}$ and $\hat{T}^2=-(t_3^L)^2=\frac{1}{4}$, and $\mathrm{Tr}\,P_L=32$; $t_3^LP_R=0$. The electric charges of the left doublets sum to zero. □

The charge that the second-rung construction gauges, $Q=\frac{1}{2}J_1$, therefore becomes at $n=3$ a universal charge on the P_* matter, and with $\mathfrak{su}(2)_L$ it cannot be gauged on that matter. The charge for which Π_A plays the role of [1, Def. 26.5] with electromagnetic content is B_{em}. Both meet the massless criterion of [1, Cor. 26.2] at Π_A (Theorem 1.13). Which of them is the photon is an identification and is not decided by the geometry of F_3 (open).

1.4 Kinetic term, current and transport on F_3

Lemma 1.12 (Vector model; fixed)**.** *For* $P\in\widetilde{\mathrm{Gr}}_4(V_+)$ *and* $X,Y\in\mathfrak{k}$,

$$\mathrm{Tr}([X,\Pi_P][Y,\Pi_P])=64\,\mathrm{tr}_8([\rho_+X,p_P][\rho_+Y,p_P]).$$

Hence on F_3: $\frac{f^2}{2}\mathrm{Tr}(D\Pi D\Pi)=32f^2\,\mathrm{tr}_8(Dp\,Dp)$; *for* $Q\in\mathfrak{k}$, $j_\Pi=-f^2\,\mathrm{Tr}([Q,\Pi]D\Pi)=-64f^2\,\mathrm{tr}_8([\rho_+Q,p]Dp)$; *and*

$$V_{\mathfrak{h}}(\Pi_P)=64\sum_a g_a^2\,\|[\rho_+T_a,p_P]\|^2,$$

in which the $\mathfrak{so}(6)$ *components of the* T_a *drop out (p1-29, p1-30).*

Proof. Both sides are bilinear in (X, Y) and invariant under K, because $\mathrm{Ad}(g)\Pi_P = \Pi_{\rho(g)P}$, $p_{RP} = Rp_PR^\top$ and ρ is equivariant. So it suffices to take $P = A$. Write $X = X_0 + \sum x_{aw}\frac{1}{2}aw$ with X_0 in the stabiliser; then $[X, \Pi_A] = \sum x_{aw}T_{aw}$, and the left side is $128\sum x_{aw}y_{aw}$. On the right, $[\rho(\frac{1}{2}aw), p_A] = -(E_{aw} + E_{wa})$, and ρ_+X_0 commutes with p_A, so the right side is $64 \cdot 2\sum x_{aw}y_{aw}$. Every tangent vector at Π_P is $[Y, \Pi_P]$ for some $Y \in \mathfrak{k}$, which gives the kinetic and current forms. □

The factor $64 = 512/8$ is the real trace on $\mathbb{R}^{512}$ against that on $\mathbb{R}^8$. At $n = 2$ the same factor is $8 = 64/8$. With $\mathrm{Tr}(T_iT_j) = 16\delta_{ij}$ there, this reproduces the $8f^2$ and $16f^2$ of [1, Thm. 26.7].

Theorem 1.13 (Kinetic term and current at Π_A; fixed)**.** *With $\Pi = \mathrm{Ad}(e^X)\Pi_A$, $X = \sum x_{aw}\frac{1}{2}aw$, and the conventions of [1, Thm. 26.7], to quadratic order in x:*

$$\mathcal{L}_\Pi = 64f^2\sum_{a,w} D_\mu x_{aw}D^\mu x_{aw}, \qquad \begin{aligned} j^\mu_{\Pi,A} &= 128f^2\sum_{w\in W}\mathrm{Im}(\bar\phi_w D^\mu\phi_w),\\ j^\mu_{\Pi,\mathrm{em}} &= 128f^2\sum_{a\in A}\mathrm{Im}(\bar\phi_a D^\mu\phi_a),\end{aligned}$$

with $\phi_w = x_{2,w} + ix_{3,w}$ and $\phi_a = x_{a,13} + ix_{a,7}$. Since $[Q, \Pi_A] = 0$ for both charges, no $A_\mu A^\mu$ term arises at $x = 0$, and the criterion of [1, Cor. 26.2] holds at Π_A and Π_W for $\mathfrak{u}(1)_A$ and for $\mathfrak{u}(1)_{\mathrm{em}}$. On the Clifford spheres of Proposition 1.6(iii) the formulas are exact, with $\mathcal{L}_\Pi = 64f^2\,Dn\cdot Dn$ (p1-31, p1-32).

Proof. $D\Pi = [DX, \Pi_A] + O(x^2)$, and j_Π at second order is $-f^2\,\mathrm{Tr}([[Q, X], \Pi_A][DX, \Pi_A])$. p1-31 evaluates both bilinear forms with $\mathrm{Tr}(T_{aw}T_{a'w'}) = 128\delta$ and the pairs of Theorem 1.10. The sign pattern is that of [1, Thm. 26.7]: $[Q, T_{\mathrm{re}}] = -T_{\mathrm{im}}$ and $[Q, T_{\mathrm{im}}] = T_{\mathrm{re}}$. □

Theorem 1.14 (Curved extension; fixed)**.** *(i) Let $\Omega = \Omega_{\mathrm{Lorentz}} + A^aT_a$ with T_a spanning a subalgebra $\mathfrak{s} \subset \mathfrak{k}$. Then $D\Pi_P = 0$ and $[\mathcal{R}, \Pi_P] = 0$ hold for arbitrary spacetime and gauge curvature iff $\mathfrak{s} \subset \mathfrak{so}(P)\oplus\mathfrak{so}(P^\perp\cap V_+)\oplus\mathfrak{so}(6)_C$. At Π_A and Π_W this includes $\mathfrak{u}(1)_A$, $\mathfrak{u}(1)_{\mathrm{em}}$, $\mathfrak{h}_{\mathrm{PS}}$ and $\mathfrak{h}_{\mathrm{SM}}$ (p1-35).*

(ii) At every point of F_3, if $D_\mu\Pi = [Y_\mu, \Pi]$ with $Y_\mu \in \mathfrak{k}$, the adapted connection of [1, Prop. 25.2] is $\mathcal{C}_\mu = [\Pi, D_\mu\Pi] = -\frac{1}{2}(Y_\mu - \mathrm{vol}_P Y_\mu \mathrm{vol}_P)$. This is a compact bivector of Λ^2V_+, so the adapted connection is spin-valued across the whole Grassmannian family (p1-33).

Proof. (i) The six interior bivectors commute with every Π_P and with $\mathfrak{k}$. So $D\Pi_P = A^a[T_a, \Pi_P]$ and $\mathcal{R} = \mathcal{R}_{\mathrm{Lorentz}} + F^aT_a$, and both vanish for arbitrary A and F iff each T_a is in the stabiliser of Theorem 1.5. (ii) With $v = \mathrm{vol}_P$ and $v^2 = \mathbf{1}$, $[\Pi, [Y, \Pi]] = \frac{1}{4}[v, [Y, v]] = \frac{1}{2}(vYv - Y)$. $\mathrm{Ad}(v)$ acts on vectors as -1 on P and $+1$ on $P^\perp \cap V_+$ and V_-. So it maps Λ^2V_+ and Λ^2V_- to themselves, and the Λ^2V_- part of Y cancels. □

Proposition 1.15 (Invariant couplings are constant; fixed)**.** *Every K-invariant function on F_3 is constant. In particular the full alignment energy satisfies $\frac{1}{512}\sum\|[e_ie_j, \Pi]\|^2 = 16$ on F_3, with the sum over the 43 compact blades. Every spectrum-only potential is constant on F_3. For a subalgebra $\mathfrak{h} \subsetneq \mathfrak{k}$ and positive couplings, the zero set of $V_\mathfrak{h}$ on F_3 is the fixed set $\mathrm{Fix}(\mathfrak{h}) = \{\Pi \in F_3 : [\mathfrak{h}, \Pi] = 0\}$ (p1-34, p1-42).*

Proof. F_3 is one K-orbit. The 43 blades are orthogonal of equal norm in the trace form, which $\mathrm{Ad}(K)$ preserves, so $\sum_a\|[T_a, \mathrm{Ad}(g)\Pi]\|^2 = \sum_a\|[\mathrm{Ad}(g^{-1})T_a, \Pi]\|^2$ is a sum over another orthonormal basis and has the same value. The value 16 is sixteen movers, each with $\|[e_ie_j, \Pi_A]\|^2 = 512$. $V_\mathfrak{h}$ is a sum of squares with positive weights. □

A coupling that selects a point of F_3 must therefore break K, and the natural candidates are the $V_\mathfrak{h}$ of proper subalgebras [6, 7]. Proposition 1.7 adds that every such $V_\mathfrak{h}$ is also invariant under D and under $\Pi \mapsto \mathbf{1} - \Pi$.

1.5 Fixed sets and the colour condition

Write $U_A = \mathrm{span}(E_2, E_3)$, $U_{\mathrm{em}} = \mathrm{span}(w_1, w_2)$ and $W_0 = \mathrm{span}(w_3, w_4)$.

Theorem 1.16 (Fixed sets in F_3; fixed)**.**

algebra	*fixed set in* F_3	*dimensions*	*contains*
$\mathfrak{so}(6)_C$, $\mathfrak{su}(3)_C$	*all of* F_3	16	*all points*
$\mathfrak{u}(1)_A$	$\{P \supset U_A\} \sqcup \{P \perp U_A\} \cong \widetilde{\mathrm{Gr}}_2(\mathbb{R}^6) \sqcup \widetilde{\mathrm{Gr}}_4(\mathbb{R}^6)$	8, 8	$\Pi_{\pm A}$; $\Pi_{\pm W}$
$\mathfrak{u}(1)_{\mathrm{em}}$	$\{P \supset U_{\mathrm{em}}\} \sqcup \{P \perp U_{\mathrm{em}}\} \cong \widetilde{\mathrm{Gr}}_2(\mathbb{R}^6) \sqcup \widetilde{\mathrm{Gr}}_4(\mathbb{R}^6)$	8, 8	$\Pi_{\pm W}$; $\Pi_{\pm A}$
$\mathbb{R}t_Y$	$\{\Pi_{\pm A}\} \sqcup \{\Pi_{\pm W}\} \sqcup \{A' \oplus W'\}$	0, 0, 6	
$\mathfrak{su}(2)_L$, $\mathfrak{h}_{\mathrm{SM}}$, $\mathfrak{h}_{\mathrm{PS}}$	$\{\Pi_A, \Pi_{-A}, \Pi_W, \Pi_{-W}\}$	0	*four isolated points*
$\mathfrak{k}$	$\emptyset$		

In the t_Y row, A' is a two-plane in A and W' a J_Y-complex line in W. That component is $\widetilde{\mathrm{Gr}}_2(A) \times \mathbb{C}P^1$, connected (p1-36 to p1-39).

Proof. For a connected group, Π_P is fixed iff P is invariant under $\rho_+(\mathfrak{h})$ (Lemma 1.3). Let j be a complex structure on a subspace U that kills $U^\perp$. If P is j-invariant and $v = v_U + v_\perp \in P$, then $j^2 v = -v_U \in P$, so $P = (P \cap U) \oplus (P \cap U^\perp)$ with $P \cap U$ itself j-invariant. For $U = U_A$ or U_{em}, $P \cap U \in \{0, U\}$, which gives the two Grassmannians of $U^\perp \cong \mathbb{R}^6$, each connected of dimension 8. For $U = W$ and $j = J_Y$, $P \cap W$ is a complex subspace of dimension 0, 2 or 4. For $\mathfrak{su}(2)_L$: A is trivial and W is irreducible without invariant vectors (p1-37). So $P \cap W \in \{0, W\}$, and if $P \cap W = 0$ the W-component of every vector of P is invariant, hence zero; thus $P \in \{A, W\}$. $\mathfrak{h}_{\mathrm{SM}}$ and $\mathfrak{h}_{\mathrm{PS}}$ contain $\mathfrak{su}(2)_L$ and fix A and W. $\mathfrak{so}(8)$ is irreducible on V_+. The tangent kernels in p1-36, p1-38 and p1-39 agree with these dimensions, and p1-38 shows that the four points are isolated. □

Proposition 1.17 (Unbroken algebras across F_3; fixed)**.** *The stabiliser of Π_P in $\mathfrak{h}_{\mathrm{PS}}$ is $\mathfrak{so}(6)_C \oplus \mathfrak{so}(P \cap W) \oplus \mathfrak{so}(P^\perp \cap W)$. In $\mathfrak{h}_{\mathrm{SM}}$ it is $\mathfrak{su}(3)_C \oplus (\mathfrak{u}_W \cap (\mathfrak{so}(P \cap W) \oplus \mathfrak{so}(P^\perp \cap W)))$, with $\mathfrak{u}_W = \rho(\mathfrak{su}(2)_L \oplus \mathbb{R}t_Y) \cong \mathfrak{u}(2)$. At random points the dimensions in $(\mathfrak{h}_{\mathrm{SM}}, \mathfrak{h}_{\mathrm{PS}})$ are $(9, 16)$ on both $\mathfrak{u}(1)_{\mathrm{em}}$-fixed components, namely $\mathfrak{su}(3)_C \oplus \mathfrak{u}(1)_{\mathrm{em}}$ and $\mathfrak{so}(6)_C \oplus \mathfrak{u}(1)_{\mathrm{em}}$. They are $(8, 16)$ on both $\mathfrak{u}(1)_A$-fixed components, $(8, 15)$ at generic points of F_3 and $(12, 21)$ at $\Pi_{\pm A}, \Pi_{\pm W}$. On the six-dimensional t_Y-component the stabiliser in $\mathfrak{h}_{\mathrm{SM}}$ has dimension 10 (p1-39, p1-40).*

Proof. If $X \in \mathfrak{so}(W)$ preserves P and kills A, then $X(P) \subset P \cap W$ and $X(P^\perp) \subset P^\perp \cap W$. So X has image in $(P \cap W) \oplus (P^\perp \cap W)$, and by antisymmetry it lies in $\mathfrak{so}(P \cap W) \oplus \mathfrak{so}(P^\perp \cap W)$; the converse is immediate. On $\{P \perp U_{\mathrm{em}}\}$ one generically has $P \cap W = 0$ and $P^\perp \cap W = U_{\mathrm{em}}$, and $\mathfrak{so}(U_{\mathrm{em}}) = \mathbb{R}\rho(\tau_3) = \mathbb{R}\rho(B_{\mathrm{em}}) \subset \mathfrak{u}_W$. p1-40 computes the dimensions of the intersections and of the stabilisers. □

The $\mathfrak{u}(1)_{\mathrm{em}}$-fixed component through Π_A is $\widetilde{\mathrm{Gr}}_4(A \oplus W_0)$. Its tangent at Π_A is $A \otimes W_0$, the neutral lower components of the four doublets of Theorem 1.10. Along it $\mathfrak{su}(3) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$ breaks to $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{\mathrm{em}}$, the pattern of a neutral doublet value (interpretive).

Corollary 1.18 (Critical points and Hessians; fixed)**.** *Every smooth function on F_3 invariant under the connected group of $\mathfrak{h} \in \{\mathfrak{su}(2)_L, \mathfrak{h}_{\mathrm{SM}}, \mathfrak{h}_{\mathrm{PS}}\}$ has $\Pi_{\pm A}$ and $\Pi_{\pm W}$ as critical points. Its Hessian there lies in the space of invariant symmetric forms on $A \otimes W$. For $\mathfrak{h}_{\mathrm{PS}}$ this space has dimension 10, the forms $S \otimes \delta_W$ with S symmetric on A. For $\mathfrak{h}_{\mathrm{SM}}$ it has dimension 16, the forms $S \otimes \delta_W + N \otimes \omega_Y$ with N antisymmetric and $\omega_Y(w, w') = \langle w, J_Y w' \rangle$ (p1-41).*

Proof. At a fixed point the differential is an invariant covector, and by p1-38 there is none other than zero [4, 5]. The Hessian at a critical point is invariant. The counts follow from Schur's lemma, with the commutant $\mathbb{R}$ of $\mathfrak{so}(4)_W$ on W and the commutant $\mathbb{C} = \mathrm{span}(\mathbf{1}, J_Y)$ of $\mathfrak{u}_W$; p1-41 computes them. □

Remark 1.19 (The fourth rung, in dimensions; fixed)**.** At $n = 4$ the internal signature is $(11, 9)$. The stabiliser of $E_2E_3E_8E_9$ in $\mathfrak{so}(11, 9)$ has dimension 126 with compact part 63, the $n = 4$ row of [1, Prop. 26.9]. The family $F_4 = \widetilde{\mathrm{Gr}}_4(\mathbb{R}^{11})$ has dimension $91 - 63 = 28$ (p1-43). All 36 bivectors of V_- commute with every point, so any colour algebra in $\mathfrak{so}(V_-)$ passes the colour test everywhere. At Π_A a $\mathfrak{u}(1)$ on a plane inside A fixes a component of dimension $14 = \dim \widetilde{\mathrm{Gr}}_2(\mathbb{R}^9)$, and one on a plane orthogonal to A fixes one of dimension $20 = \dim \widetilde{\mathrm{Gr}}_4(\mathbb{R}^9)$. $\mathfrak{so}(4)_W$ fixes a component of dimension $12 = \dim \widetilde{\mathrm{Gr}}_4(\mathbb{R}^7)$ through Π_A, so at $n = 4$ the Pati–Salam fixed points are no longer isolated (p1-44). In general $\dim F_n = 4(3n - 5)$, and a $\mathfrak{u}(1)$ on one positive plane fixes components of dimensions $2(3n-5)$ and $4(3n-7)$: (2), (8, 8), (14, 20) at $n = 2, 3, 4$ (p1-45).

1.6 The answer

(a) *The family.* The $n \geq 3$ analogue of the compact family is $F_n = \{\Pi_P : P \in \widetilde{\mathrm{Gr}}_4(V_+)\}$, the K_n-orbit of Π_A; at $n = 2$ it is the paper's S^4 (fixed). At $n = 3$ it is $\widetilde{\mathrm{Gr}}_4(\mathbb{R}^8)$, of dimension 16, a compact symmetric space [3]. Its metric is 64 times the trace metric of the Grassmannian (Lemma 1.12). The paper's S^4 is one of the inner Clifford spheres in it, and is not K-invariant (fixed).

(b) *Colour.* Every point of F_3 commutes with all of $\mathfrak{so}(6)_C \supset \mathfrak{su}(3)_C$, and so does every point of F_n with $\mathfrak{so}(V_-)$. The other involutive four-blade types never commute with $\mathfrak{su}(3)_C$ (fixed).

(c) *The vacuum.* Π_A is the analogue of [1, Def. 26.5]. It is Lorentz invariant, neutral under $\mathfrak{u}(1)_A$ and $\mathfrak{u}(1)_{\mathrm{em}}$, fixed by $\mathfrak{h}_{\mathrm{PS}}$ and $\mathfrak{h}_{\mathrm{SM}}$, moved by 16 compact bivectors, and parallel under any connection valued in its stabiliser. It is the point $n = c$ of the embedded second-rung family. The literal point $n = e$ is not neutral at $n = 3$ (fixed).

(d) *Data for a selecting coupling.* A coupling that distinguishes the points of F_3 is not K-invariant (obstructed for K-invariant couplings). Its zero set, if it is an alignment energy, is one of the fixed sets of Theorem 1.16. Its Hessian at the four Pati–Salam points lies in the 10- or 16-dimensional spaces of Corollary 1.18. Its fluctuations there are four doublets $(\mathbf{1}, \mathbf{2})_{1/2}$ with a family index (fixed). No alignment energy of a subalgebra of $\mathfrak{k}$ separates Π_A, Π_{-A}, Π_W and Π_{-W}; a coupling to the chiral block does, since on the block D exchanges P_L and P_R (fixed). This is the counterpart on F_3 of [1, Prop. 21.6] (interpretive).

(e) *The charge.* On P_* the construction's charge $\frac{1}{2}J_1$ is $\frac{1}{2}J$, universal, and anomalous together with $\mathfrak{su}(2)_L$ (fixed; restated). The electromagnetic charge B_{em} also leaves Π_A massless and has four charged complex fluctuations. Which charge, which coupling and which scale select the vacuum is not decided here (open).

2 Which charge is the photon, and the residual choice

The second-rung construction gauges the charge $Q = \frac{1}{2}J_1$ and carries it, in the curved extension, by the normal $SO(2)$ connection [1, Thms. 26.7, 26.8]. At the third rung that charge is $Q_A = \frac{1}{2}q_A = \frac{1}{2}E_2E_3$, and the matter block of [1, Section 8] carries a second candidate, the electric charge B_{em}. This section decides which of the two is the photon on the family F_3 of Section 1, and what is left to choose once a gauged subalgebra has aligned the vacuum. The results are as follows. The charge operator of Q_A is half the involution U that cuts out P_*, so on the block Q_A equals $\frac{1}{2}$ on every state: it is half the fermion number, and it is anomalous together with $\mathfrak{su}(2)_L$ (fixed). It is not electromagnetism on that matter, whatever the vacuum (obstructed). The electric charge leaves the four aligned vacua massless, has four charged complex fluctuations with the electron's charge magnitude, and acts on the block with the Standard-Model charges and without anomaly. With B_{em} gauged, the massless vector of the projector current is the

statement	class	checks
Proposition 1.1: second-rung data, K_2-orbit $= S^4 = \widetilde{\mathrm{Gr}}_4(\mathbb{R}^5)$, Q-fixed S^2, constant potentials	restated (ES17 Thms 26.6–26.8, Prop. 26.9); orbit and fixed set fixed	p1-1 to p1-7
Lemma 1.2: four-blade census, even-grade rule, $\mathrm{vol}_+ = -Z$	fixed	p1-8 to p1-10
Lemma 1.3: $\mathrm{vol}_P^\top = \mathrm{vol}_{\theta P}$, recovery of P	fixed	p1-13
Lemma 1.4: $\mathfrak{su}(3)_C$ irreducible on V_-	fixed	p1-15
Theorem 1.5: three K-orbits, stabilisers, spectra, colour test	fixed; $n = 3$ row of ES17 Prop. 26.9 restated	p1-11, p1-12, p1-14, p1-16, p1-42
Proposition 1.6: Clifford spheres, embedded second-rung family	fixed	p1-19, p1-32, p1-47
Proposition 1.7: duality $P \mapsto P^\perp$, invariance of mass matrices	fixed	p1-20, p1-46
Theorem 1.9: Π_A, Π_W neutral, $\mathfrak{h}_{\mathrm{PS}}$- and $\mathfrak{h}_{\mathrm{SM}}$-fixed, not rigid; literal embedding fails	fixed	p1-21 to p1-23, p1-35
Theorem 1.10: fluctuations, four doublets $(\mathbf{1},\mathbf{2})_{1/2}$, charges	fixed	p1-24 to p1-26
Proposition 1.11: $\Pi_A P_* = P_L$, $q_A = J$ on the block, $P_* T P_* = 0$, anomaly	fixed; (iv) restated (ES17 Prop. 8.5); anomaly condition inherited	p1-17, p1-18, p1-27, p1-28
Lemma 1.12: vector model, kinetic term, current, $V_{\mathfrak{h}}$	fixed	p1-29, p1-30
Theorem 1.13: $\mathcal{L}_\Pi$ and j_Π at Π_A and on spheres	fixed	p1-31, p1-32
Theorem 1.14: parallel vacua, compact adapted connection	fixed	p1-33, p1-35
Proposition 1.15: K-invariant couplings constant; zero sets	fixed; selection by K-invariant couplings obstructed	p1-34, p1-42
Theorem 1.16, Proposition 1.17: fixed sets, unbroken algebras	fixed	p1-36 to p1-40
Corollary 1.18: critical points and Hessians	fixed; criticality principle inherited	p1-38, p1-41
Remark 1.19: fourth rung	fixed; $n = 4$ row of ES17 Prop. 26.9 restated	p1-43 to p1-45
family reading of $\mathfrak{so}(4)_A$; electroweak pattern on the neutral component	interpretive	—
which charge and coupling select the vacuum	open	—

Table 1: Claims of Section 1 and their classes. The checks are the `PASS` lines printed by `verify/p1_families.py`.

electromagnetic photon of the block, and the normal $U(1)$ is a separate abelian field (fixed). Of the three counts against the normal $U(1)$ in [1, Section 31] only the sign count changes status, and a fourth count, the charge on the block, decides. After alignment by $\mathfrak{h}_{\mathrm{SM}}$ four vacua remain. A Klein group of symmetries of the projector sector permutes them simply transitively, so no alignment energy, kinetic term or current separates them (fixed). The overlaps with P_*, with the lepton projector and with the chirality halves do not separate them either (fixed). The overlap with the half P_L of the block on which $\mathfrak{su}(2)_L$ acts selects the pair $\{\Pi_A, \Pi_{-W}\}$, and the two members of the pair coincide on the half of the module that contains the block (fixed). A single point needs a coupling that sees the other half, for instance the overlap with P_{fam} (constructed). Identifiers p3-n refer to the `PASS` lines of `verify/p3_identification.py` (Table 5).

Conventions. Definitions and notation are those of Section 1; in particular F_3, $\Pi_{\pm A}$, $\Pi_{\pm W}$, Q_A, B_{em}, $t_i^{L,R}$, t_Y, H_C, ρ_+ and vol_+ are as there. For a compact generator X commuting with J the charge operator is $\hat{X} := -JX$. It is symmetric, and $X = J\hat{X}$ acts as $i\hat{X}$ in the complex structure J [1, Section 8]. The inner product on $\mathfrak{k}$ is $\langle X, Y\rangle = -\mathrm{Tr}(XY)/512$. For it the unit blades $e_i e_j$ have norm one, $|t_3^L|^2 = \frac{1}{8}$ and $|t_Y|^2 = \frac{5}{24}$. Orthonormal bases are $2\sqrt{2}\, t_i^L$, $\sqrt{24/5}\, t_Y$ and an orthonormal basis of $\mathfrak{su}(3)_C$ for $\mathfrak{h}_{\mathrm{SM}}$, with couplings g_2, g_Y, g_3, and the unit blades of $\mathfrak{so}(4)_W \oplus \mathfrak{so}(6)_C$ for $\mathfrak{h}_{\mathrm{PS}}$, with one coupling g. We write $U_A = \mathrm{span}(E_2, E_3)$, $A_0 = \mathrm{span}(E_8, E_9)$, $U_{\mathrm{em}} = \mathrm{span}(E_7, E_{13})$, $W_0 = \mathrm{span}(E_{14}, E_{15})$ and $P_{L,R} = P_* \frac{1}{2}(\mathbf{1} \mp \Gamma)$. For an oriented four-

field	Γ	$\hat{T}_L^3$	$\hat{Y}$	$\hat{Q}_{\mathrm{em}}$	$\hat{Q}_A$	$B-L$	real dim.	complex dim.
Q_L	-1	$\pm\frac{1}{2}$	$\frac{1}{6}$	$\frac{2}{3},\ -\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	$12+12$	$6+6$
L_L	-1	$\pm\frac{1}{2}$	$-\frac{1}{2}$	$0,\ -1$	$\frac{1}{2}$	-1	$4+4$	$2+2$
u_R	$+1$	0	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	12	6
d_R	$+1$	0	$-\frac{1}{3}$	$-\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	12	6
e_R	$+1$	0	-1	-1	$\frac{1}{2}$	-1	4	2
ν_R	$+1$	0	0	0	$\frac{1}{2}$	-1	4	2

Table 2: Joint eigenvalues on V_* (p3-2). Complex dimensions refer to the complex structure J; one internal Weyl component has four real, or two complex, dimensions. The columns $\hat{Y}$ and $\hat{Q}_{\mathrm{em}}$ restate [1, Prop. 8.4]. The column $\hat{Q}_A$ is constant.

plane $P \subset V_+$ with oriented orthonormal frame $(f_1, \dots, f_4)$, written as a 4×8 matrix F in the generator basis of V_+, the Plücker coordinates are $p_I(P) = \det F_I$ for the 70 four-subsets I of that basis, and e_I is the blade of I in increasing order. We write p_A and p_W for $I = A$ and $I = W$. The family projector is $P_{\mathrm{fam}} = \frac{1}{8}\prod_i(\mathbf{1} + X_i)$ [1, Section 7], and $Q_v = P_L + \gamma^0 P_L \gamma^0$ [1, Prop. 21.6].

Definition 2.1 (The lepton projector)**.** P_{lep} is the spectral projector of the colour charge $Q_c = -JH_C$ for the eigenvalue -3. With $s_k = J\,E_{9+k}E_{15+k}$, $k = 1, 2, 3$, one has $Q_c = s_1 + s_2 + s_3$ and $P_{\mathrm{lep}} = \frac{1}{8}\prod_k(\mathbf{1} - s_k)$.

On P_* the eigenvalue -3 of Q_c is the colour singlet [1, Prop. 8.4]. Proposition 2.18 shows that P_{lep} reproduces the row of [1, Prop. 21.6], where the colour-singlet lepton projector is named but not written out.

2.1 Two charges on the matter

Proposition 2.2 (The construction's charge is half of U; fixed, and restated)**.** *(i) $\hat{Q}_A = -JQ_A = \frac{1}{2}U$ on $\mathbb{R}^{512}$, where $U = -Jq_A$ is the third involution of P_*. Since $UP_* = P_*$ and q_A commutes with X, Z and U,*

$$q_A P_* = JP_*, \qquad \hat{Q}_A P_* = \tfrac{1}{2}P_*.$$

The first identity is the statement $q_A|_{V_*} = J$ of [1, Section 8] (p3-1).

(ii) On $V_* = \operatorname{im} P_*$ the operators Γ, $\hat{T}_L^3 = -Jt_3^L$, $\hat{Y} = -Jt_Y$, $\hat{Q}_{\mathrm{em}} = -JB_{\mathrm{em}}$, $\hat{Q}_A$ and $B-L = -JH_C/3$ commute. Their joint spectrum is that of Table 2 (p3-2).

(iii) On the Weyl block $\frac{1}{2}(\mathbf{1} - \Gamma)$ of the spectator action, of rank 256, $\hat{Q}_A$ takes the values $\pm\frac{1}{2}$, each 128 times. There $\hat{Q}_{\mathrm{em}}$ takes the values ±1 (16 times each), $\pm\frac{2}{3}$ (48), $\pm\frac{1}{3}$ (48) and 0 (32), the charges of the two families and two mirror families of [1, Prop. 10.12] (p3-3).

Proof. (i) $\hat{Q}_A = -J\frac{1}{2}q_A = \frac{1}{2}U$. As index sets, $X \sim I \cup A \cup C$ with I the interior, $Z \sim A \cup W$ and $U \sim \{8, 9\} \cup W \cup C$. The bivector q_A shares two indices with X and with Z and none with U, so by equation (3) of [1] it commutes with all three. X, Z, U commute, so $UP_* = P_*$, that is $-Jq_AP_* = P_*$; multiplying by J and using $J^2 = -\mathbf{1}$ gives $q_AP_* = JP_*$. (ii) The operators are the charge operators of commuting compact elements, together with Γ, which commutes with every internal bivector; p3-2 diagonalises them jointly on V_*. (iii) U commutes with Γ and is traceless on the Weyl block; p3-3 computes both spectra there. □

Corollary 2.3 (What the construction's photon couples to; fixed, restated and obstructed)**.** *(i) A gauge field coupled through Q_A couples to every Weyl fermion of the block with the same charge $\frac{1}{2}$, on both chiralities. In the complex structure J each of the sixteen Weyl fermions*

has fermion number one [1, Section 8]. The coupling is therefore to $\frac{1}{2}F$, $F = 3B + L$, *a universal vector charge (fixed).*

(ii) No real λ *gives* $\hat{Q}_A P_* = \lambda \hat{Q}_{\mathrm{em}} P_*$, *since* $\hat{Q}_{\mathrm{em}}$ *vanishes on the neutrinos and* $\hat{Q}_A$ *does not (p3-4). The statement involves only* P_*, *so the construction's charge is not the electric charge on the block for any vacuum projector (obstructed).*

(iii) $\hat{Q}_{\mathrm{em}}$ *has the same spectrum on* P_L *as on* P_R, *so its cubic, gravitational and mixed colour anomalies cancel between the chiralities, and* $\mathrm{Tr}(P_L \hat{Q}_{\mathrm{em}} (\hat{T}^3_L)^2) = 0$ *(fixed; the anomaly conditions are inherited [8, 9]). For* $\hat{Q}_A$ *the same traces vanish except* $\mathrm{Tr}(P_L \hat{Q}_A (\hat{T}^3_L)^2) = 4$. *This is the mixed* $SU(2)^2_L U(1)_A$ *anomaly of [1, Prop. 8.5] at* $a = \frac{1}{2}$, $b = 0$ *(restated; p1-28, p3-5). With* P_* *as the matter and* $\mathfrak{su}(2)_L$ *gauged,* $\mathfrak{u}(1)_A$ *cannot be gauged (obstructed under these hypotheses).*

Proof. (i) and (ii) read off Table 2. (iii) is p3-5. The anomaly of $\hat{Q}_A$ is that of fermion number in the Standard Model, which $SU(2)_L$ instantons violate [10]. □

The universality of Q_A on matter is therefore not a property of the vacuum. It is the factor $\frac{1}{2}(\mathbf{1} + U)$ of P_*: the block is cut out as an eigenspace of the construction's charge (Proposition 2.2(i)), and on the rest of the spectator Weyl block the same charge takes the value $-\frac{1}{2}$ (Proposition 2.2(iii)). At the second rung the coupled model and the vacuum share the charge $\frac{1}{2}J_1$ [1, Section 26]. At the third rung that shared charge is half the fermion number.

2.2 The counts against the normal $U(1)$, and the dichotomy

Table 3 takes the three counts of [1, Section 31] to the third rung and adds the count of Corollary 2.3. The four aligned vacua $\Pi_0 \in \{\Pi_{\pm A}, \Pi_{\pm W}\}$ commute with both Q_A and B_{em} (Theorem 1.9), so both charges can be compared at the same vacuum.

Row (iii) rests on [1, Cor. 23.6, Thm. 30.4]: there a vector obtained by reduction on a plane of the sign of time has a kinetic term of the wrong sign [11]. That theorem is about a metric origin of the field. An internal gauge field with an adopted Maxwell term is not covered by it. For a compact generator the invariant form is positive ([1, Thm. 2.3], Table 3), so a Maxwell term of the healthy sign exists, as in ordinary Yang–Mills theory. If $\mathfrak{u}(1)_{\mathrm{em}}$ were itself obtained by reduction, its weak part would rotate the plane U_{em}, of the sign of time, and the same mechanism would apply to that part and not to the colour part $\frac{1}{6}H_C$ (interpretive). The same holds for Q_A, whose plane U_A also has the sign of time. Count (iii) therefore separates the two fields only by their origin, and not by their generator.

The dichotomy at the end of [1, Section 25] reads: either the normal $U(1)$ is not electromagnetic, or the paired-mode projector is not the observation projector. [1, Section 26] resolves it in one direction: if the normal $U(1)$ is to be electromagnetism, the paired-mode projector must go. At the third rung count (iv) decides the first alternative for every projector, because it is a property of P_* through its factor U: on the block matter the normal $U(1)$ is not electromagnetic (obstructed). The replacement of the paired-mode projector is still available, as Π_A or any point of the aligned set, and the charge that uses it is B_{em}. At $n = 3$ both alternatives of the dichotomy hold, and neither is a choice.

2.3 Identification at the aligned vacua

Proposition 2.4 (Joint fixed points of the two charges; fixed)**.** *The points of* F_3 *commuting with both* Q_A *and* B_{em} *form six components: the isolated points* $\Pi_{\pm(U_A \oplus U_{\mathrm{em}})}$, *the isolated points* $\Pi_{\pm R}$ *with* $R = \mathrm{span}(E_8, E_9, E_{14}, E_{15})$, *and two copies of* $\widetilde{\mathrm{Gr}}_2(\mathbb{R}^4)$, *of dimension* 4,

$$\{\Pi_{U_A \oplus P'} : P' \in \widetilde{\mathrm{Gr}}_2(R)\}, \qquad \{\Pi_{U_{\mathrm{em}} \oplus P'} : P' \in \widetilde{\mathrm{Gr}}_2(R)\}.$$

$\Pi_{\pm A}$ *lie on the first of these, with tangent* $A_0 \otimes W_0$, *and* $\Pi_{\pm W}$ *on the second (p3-6).*

count	normal $U(1)$, generator Q_A	$\mathfrak{u}(1)_{\mathrm{em}}$, generator B_{em}	change
(i) a spinless test particle cannot see it [1, Prop. 23.4, Thm. 30.1]	holds: it concerns the metric charge of the normal plane; the field couples through the spinor sector, at Π_0 to four complex scalars ϕ_w and to the block	not applicable, since B_{em} is not a metric charge; it couples through the spinor sector to four complex scalars ϕ_a and to the block, and a neutral spinless field does not see the ordinary photon either	none in substance (restated; interpretive reading)
(ii) the adapted generator does not commute with the physical generators [1, Section 25]	removed: $[Q_A, \Pi_0] = 0$, $D\Pi_0 = 0$, and the adapted correction is compact and commutes with the interior gammas [1, Thm. 26.8], Theorem 1.14	removed in the same way	none at $n = 3$; removed already at $n = 2$ (fixed)
(iii) it gravitates with the wrong sign [1, Cor. 23.6, Thm. 30.4]	holds for the Kaluza–Klein vector of the normal plane	not covered: the Maxwell term is adopted, and a positive one exists since $\langle B_{\mathrm{em}}, B_{\mathrm{em}}\rangle > 0$; read as a Kaluza–Klein vector, the plane U_{em} has the sign of time, like the normal plane	from forced to adopted (constructed); the Kaluza–Klein reading is interpretive
(iv) the charge on the block, Corollary 2.3	universal $\frac{1}{2}$, anomalous with $\mathfrak{su}(2)_L$	the Standard-Model charges, anomaly free	decides (obstructed for Q_A, fixed for B_{em})

Table 3: The counts against the normal $U(1)$ at the third rung (p3-4, p3-5). In row (iii), E_2, E_3, E_7, E_{13} square to $+\mathbf{1}$, as the time generator $\gamma^0 = E_1$ does, and the colour generators to $-\mathbf{1}$ (p3-4).

Proof. By Theorem 1.16, Π_P commutes with Q_A iff $P \supset U_A$ or $P \perp U_A$, and with B_{em} iff $P \supset U_{\mathrm{em}}$ or $P \perp U_{\mathrm{em}}$. The four combinations give $P = U_A \oplus U_{\mathrm{em}}$; $P = U_A \oplus P'$ with $P' \subset (U_A \oplus U_{\mathrm{em}})^{\perp} = R$; $P = U_{\mathrm{em}} \oplus P'$; and $P = R$. An orientation of a direct sum is a pair of orientations of the summands up to a common reversal. This gives $\widetilde{\mathrm{Gr}}_2(R)$ in the middle cases and two points in the outer ones. Finally $A = U_A \oplus A_0$ and $W = U_{\mathrm{em}} \oplus W_0$ with $A_0, W_0 \subset R$. p3-6 computes the joint tangent kernels 0, 0, 4, 4 at representatives. □

Proposition 2.5 (Charges of the fluctuations; fixed)**.** *At each of $\Pi_{\pm A}$, $\Pi_{\pm W}$ the sixteen tangent directions decompose under the torus $\mathfrak{u}(1)_A \times \mathfrak{u}(1)_{\mathrm{em}}$ into complex lines of weights*

$$U_A \otimes U_{\mathrm{em}} :\ (1,1) \oplus (1,-1), \qquad U_A \otimes W_0 :\ 2 \times (1,0), \qquad A_0 \otimes U_{\mathrm{em}} :\ 2 \times (0,1),$$

and four real neutral directions $A_0 \otimes W_0$ (p3-7). Eight real directions are charged under each $\mathfrak{u}(1)$, four under both and four under neither, in agreement with Theorem 1.10(iii),(iv).

Proof. $\rho_+(Q_A)$ is the rotation of U_A at unit rate and $\rho_+(B_{\mathrm{em}})$ the rotation of U_{em}; on $A \otimes W$ the first acts on the first factor and the second on the second factor. □

Theorem 2.6 (The photon of the aligned vacua; fixed, with the gauged generator a choice)**.** *Let $\Pi_0 \in \{\Pi_{\pm A}, \Pi_{\pm W}\}$. Let the internal connection carry a field A_μ along a compact generator $Q \in \mathfrak{k}$, acting on spinors by Q and on the projector by $\operatorname{ad} Q$, and adopt a Maxwell term.*

(a) For $Q = B_{\mathrm{em}}$ the four requirements on an electromagnetic photon of the block hold. (1) The generator is the electric charge: $\hat{Q}_{\mathrm{em}} = \hat{T}^3_L + \hat{Y}$ on V_*, with the values of Table 2. (2) $[B_{\mathrm{em}}, \Pi_0] = 0$, so the kinetic term gives no $A_\mu A^\mu$ term. (3) Charged fluctuations exist: the four complex scalars ϕ_a of Theorem 1.10(iii), with current $j^\mu_{\Pi,\mathrm{em}} = 128 f^2 \sum_a \mathrm{Im}(\bar{\phi}_a D^\mu \phi_a)$ (Theorem 1.13). (4) The matter carries the Standard-Model charges without anomaly (Corollary 2.3(iii)). Both currents have the same unit of charge: $B^2_{\mathrm{em}} = -\hat{Q}^2_{\mathrm{em}}$ on V_*,

with value -1 *on the electron, and* $(\operatorname{ad} B_{\mathrm{em}})^2 = -1$ *on the charged tangent directions (p3-8). The scalars of the projector current carry the electron's charge magnitude, and no normalisation is free.*

(b) *For* $Q = Q_A$ *requirements (2) and (3) hold and (1) and (4) fail (Corollary 2.3). The massless vector couples to* $\frac{1}{2}F$ *on the block and to four complex scalars* ϕ_w *of charge* 1*, twice the fermion charge (p3-8). It is not the electromagnetic photon (obstructed).*

(c) Q_A *and* B_{em} *commute and are orthogonal.* Q_A *commutes with* $\mathfrak{h}_{\mathrm{SM}}$ *and* $\mathfrak{h}_{\mathrm{PS}}$ *and lies in the centraliser* $\mathfrak{so}(4)_A$ *of* $\mathfrak{h}_{\mathrm{PS}}$ *in* $\mathfrak{k}$ *(p3-4). The normal* $SO(2)$ *connection of [1, Thm. 26.8], whose plane is the surplus pair* E_2, E_3 *of the first constituent, is the component of the connection along* Q_A*. It is a second abelian field, not the photon (fixed).*

(d) *All twelve generators of* $\mathfrak{h}_{\mathrm{SM}}$ *commute with* Π_0*, so with* $\mathfrak{h}_{\mathrm{SM}}$ *gauged no gauge boson is massive at* Π_0*, and* $\mathfrak{u}(1)_{\mathrm{em}}$ *is singled out only by an electroweak stage. The projector's own neutral directions provide one. At* $\Pi_\theta = \mathrm{Ad}(\exp \frac{\theta}{2} E_8 E_{14})\Pi_A$*,* $\theta \notin \pi\mathbb{Z}$*, the mass matrix of* $\mathfrak{h}_{\mathrm{SM}}$ *has the eigenvalue* 0 *nine times (gluons and photon),* m_W^2 *twice and* m_Z^2 *once, with*

$$\frac{m_W^2}{m_Z^2} = \frac{1}{1 + \frac{3}{5}\, g_Y^2/g_2^2},$$

and its massless electroweak combination is B_{em} *(p3-9). At* $g_Y = g_2$ *the ratio is* $\frac{5}{8} = 1 - \frac{3}{8}$*, the value* $\cos^2\theta_W$ *for the* $\sin^2\theta_W = \frac{3}{8}$ *of [1, Thm. 21.1].*

Proof. (a) (1) is Proposition 2.2(ii). (2) and (3) are Theorems 1.9, 1.10 and 1.13. At $\Pi_{-A} = \mathbf{1} - \Pi_A$ the tangents change sign and nothing else changes, and at $\Pi_{\pm W}$ Theorem 1.10(v) applies. For the unit: on V_*, $B_{\mathrm{em}} = J\hat{Q}_{\mathrm{em}}$ with J commuting and $J^2 = -\mathbf{1}$, so $B_{\mathrm{em}}^2 = -\hat{Q}_{\mathrm{em}}^2$; on the tangent, $[B_{\mathrm{em}}, T_{a,7}] = T_{a,13}$ and $[B_{\mathrm{em}}, T_{a,13}] = -T_{a,7}$ (Theorem 1.10). (b) p3-8 gives $(\operatorname{ad} Q_A)^2 = -1$ on $U_A \otimes W$ and $Q_A^2 = -\frac{1}{4}$ on V_*. (c) is p3-4 together with the form $\Omega = \Omega_{\mathrm{Lorentz}} + AQ$, $Q = \frac{1}{2}ab$, of the curved extension [1, Thm. 26.8]. (d) $\mathfrak{su}(3)_C$ acts trivially on V_+. In the vector model of Lemma 1.12, $p_\theta = p_{\mathrm{span}(E_2,E_3,E_9)} + uu^\top$ with $u = \cos\theta\, E_8 \pm \sin\theta\, E_{14}$. For $x \in \rho_+(\mathfrak{h}_{\mathrm{SM}})$, which kills A, $[x, uu^\top] = au^\top + ua^\top$ with $a = xu = \pm\sin\theta\, xE_{14}$, so the mass matrix is $128\sin^2\theta$ times the Gram matrix of the vectors $g_a\rho_+(T_a)E_{14}$. The vectors $\rho_+(t_i^L)E_{14}$ are orthogonal of length $\frac{1}{2}$, because the $2\rho_+(t_i^L)$ are anticommuting complex structures, and $\rho_+(t_Y)E_{14} = -\rho_+(t_3^L)E_{14}$, because B_{em} kills E_{14}. In the basis of the conventions the Gram matrix is $2g_2^2$ on the first two generators, and on $(2\sqrt{2}\,t_3^L, \sqrt{24/5}\,t_Y)$ it is $\left(\begin{smallmatrix} 2g_2^2 & -\sqrt{12/5}\,g_2 g_Y \\ -\sqrt{12/5}\,g_2 g_Y & \frac{6}{5}g_Y^2 \end{smallmatrix}\right)$, with eigenvalues 0 and $2g_2^2 + \frac{6}{5}g_Y^2$. p3-9 computes the full 12×12 matrix on $\mathbb{R}^{512}$. □

So at the third rung the photon of the aligned vacua is the $\mathfrak{u}(1)_{\mathrm{em}}$ gauge field of the internal connection, and the normal $U(1)$ is a separate abelian field. On the block that field couples to fermion number, and with $\mathfrak{su}(2)_L$ gauged it cannot be gauged on that matter (Corollary 2.3). The construction's identification of its photon with the normal connection is therefore not compatible with the matter at $n = 3$ (obstructed). What the second rung established survives with B_{em} in place of Q_A: the massless criterion of [1, Cor. 26.2] at a Lorentz-invariant vacuum, the charged-scalar current, and a curved extension in which the vacuum is parallel (Theorem 1.14). The theorem rests on the choice of the gauged generator, and B_{em} is fixed only once the Standard-Model embedding of [1, Section 8] is adopted (constructed). The coupling of the second abelian field and the values of f and of the gauge couplings are not decided (open). Theorem 2.6(d) shows that the projector can itself carry an electroweak stage: its neutral doublet directions break $\mathfrak{h}_{\mathrm{SM}}$ to $\mathfrak{su}(3)_C \oplus \mathfrak{u}(1)_{\mathrm{em}}$ with the tree-level relation of one doublet [12, 13]. Whether the vacuum moves there is decided by the potential, which Section 2.5 examines.

M	Π_A	Π_{-A}	Π_W	Π_{-W}	$\mathrm{Tr}(\Pi_P M)$ on F_3
P_*	$(32,32)$	$(32,32)$	$(32,32)$	$(32,32)$	32
P_{lep}	$(32,32)$	$(32,32)$	$(32,32)$	$(32,32)$	32
$\frac{1}{2}(\mathbf{1}\pm\Gamma)$	$(128,128)$	$(128,128)$	$(128,128)$	$(128,128)$	128
P_L	$(32,32)$	$(0,0)$	$(0,0)$	$(32,32)$	$16(1+p_A-p_W)$
P_R	$(0,0)$	$(32,32)$	$(32,32)$	$(0,0)$	$16(1-p_A+p_W)$
Q_v	$(64,64)$	$(0,0)$	$(0,0)$	$(64,64)$	$32(1+p_A-p_W)$
P_{fam}	$(64,64)$	$(0,0)$	$(32,16)$	$(32,16)$	$32(1+p_A)$

Table 4: Overlaps ($\mathrm{Tr}\,\Pi M$, $\mathrm{Tr}\,\Pi M\Pi M$) at the four aligned points, and the linear overlap on all of F_3 (p3-11, p3-12).

2.4 The residual choice

By Proposition 1.15 and Theorem 1.16, with $\mathfrak{h}_{\mathrm{SM}}$ or $\mathfrak{h}_{\mathrm{PS}}$ gauged with positive couplings, the alignment energy vanishes on F_3 exactly at Π_A, Π_{-A}, Π_W and Π_{-W}. This subsection determines what separates these four points.

Lemma 2.7 (Overlaps are Plücker functions; fixed)**.** *For every oriented four-plane $P\subset V_+$, $\mathrm{vol}_P=\sum_I p_I(P)\,e_I$, and for every operator M on $\mathbb{R}^{512}$*

$$\mathrm{Tr}(\Pi_P M)=\tfrac{1}{2}\,\mathrm{Tr}\,M+\tfrac{1}{2}\sum_I p_I(P)\,\mathrm{Tr}(e_I M).$$

The four-blade components $\mathrm{Tr}(e_I M)$ vanish for $M=P_*$, P_{lep} and $\frac{1}{2}(\mathbf{1}\pm\Gamma)$. The nonzero ones are $\mathrm{Tr}(e_A P_{\mathrm{fam}})=64$, $\mathrm{Tr}(e_A P_L)=-\mathrm{Tr}(e_W P_L)=32$, their negatives for P_R, $\mathrm{Tr}(e_A Q_v)=-\mathrm{Tr}(e_W Q_v)=64$ and $\mathrm{Tr}(e_A\Pi_A)=256$ (p3-10).

Proof. The frame vectors are orthonormal, so their Clifford product equals their exterior product, $\sum_I \det F_I\, e_I$. The blades e_I are symmetric involutions with $\mathrm{Tr}(e_I e_J)=512\,\delta_{IJ}$. For the components it suffices to list index sets. The eight products of X, Z, U have the index sets $\emptyset$, $I\cup A\cup C$, $A\cup W$, $\{8,9\}\cup W\cup C$ and their symmetric differences, none of which is a four-subset of V_+. With Γ, whose index set is all eighteen generators, the products include $\Gamma X=\chi_W$ and $\Gamma XZ=-\chi_W\mathrm{vol}_+=-\mathrm{vol}_A$, so $P_L=\frac{1}{16}(\mathbf{1}+X)(\mathbf{1}+Z)(\mathbf{1}+U)(\mathbf{1}-\Gamma)$ contains $\frac{1}{16}(\mathrm{vol}_A-\chi_W)$ and no other four-blade of V_+. The products of the s_k have index sets of sizes 12, 4 inside C, and 8 equal to $A\cup W$. Among the products of the X_i only $X_1=\mathrm{vol}_A$ is a four-blade of V_+ (Proposition 2.18). γ^0 commutes with every e_I, which doubles the components of P_L in Q_v. □

Proposition 2.8 (The overlaps; fixed)**.** *(i) On all of F_3, $\mathrm{Tr}(\Pi P_*)=\mathrm{Tr}(\Pi P_{\mathrm{lep}})=32$ and $\mathrm{Tr}(\Pi\frac{1}{2}(\mathbf{1}\pm\Gamma))=128$ are constant, while*

$$\mathrm{Tr}(\Pi_P P_{\mathrm{fam}})=32(1+p_A)=\tfrac{1}{4}\,\mathrm{Tr}(\Pi_P\Pi_A),\qquad \mathrm{Tr}(\Pi_P P_L)=16(1+p_A-p_W)=\tfrac{1}{2}\,\mathrm{Tr}(\Pi_P Q_v),$$

and $\mathrm{Tr}(\Pi_P P_R)=16(1-p_A+p_W)$ (p3-12).

(ii) $\mathrm{Tr}(\Pi P_*\Pi P_*)=\|P_*\Pi P_*\|^2\le\mathrm{Tr}(\Pi P_*)=32$, with equality iff $[\Pi,P_*]=0$. The points of F_3 commuting with P_* are $\Pi_{\pm A}$ and $\Pi_{\pm W}$, which are isolated, and the two components $\{\Pi_{U_A\oplus W'}\}$ and $\{\Pi_{A_0\oplus W'}\}$, $W'\in\widetilde{\mathrm{Gr}}_2(W)$, of dimension 4 (p3-12, p3-13).

(iii) At the four points the linear and quadratic overlaps are those of Table 4 (p3-11).

Proof. (i) is Lemma 2.7 with $\mathrm{Tr}\,M$ and the components listed there. (ii) P_* is a contraction, so $\|P_*\Pi P_*\|^2\le\|\Pi P_*\|^2=\mathrm{Tr}(P_*\Pi P_*)=32$, with equality iff $(\mathbf{1}-P_*)\Pi P_*=0$, that is iff $\Pi P_*=P_*\Pi P_*$ is symmetric, that is iff Π commutes with P_*. Every vol_P commutes with J, Γ and $\mathrm{vol}_+=-Z$, so Π_P commutes with P_* iff vol_P commutes with χ_W and with q_A. $\mathrm{Ad}(\chi_W)$

acts on V_+ as the reflection -1 on W, $+1$ on A. It fixes vol_P iff $P = (P \cap A) \oplus (P \cap W)$ with $\dim(P \cap W)$ even. With Theorem 1.16 for q_A this leaves $P = \pm A$, $P = \pm W$, and $P = A' \oplus W'$ with $A' \in \{U_A, A_0\}$ and W' a two-plane in W. p3-13 checks that the four points are isolated and that the components have dimension 4. (iii) is the computation p3-11. □

Lemma 2.9 (A comass inequality; fixed)**.** *For every oriented four-plane $P \subset V_+ = A \oplus W$, $|p_A(P)| + |p_W(P)| \le 1$, with equality iff $P \in \{\pm A, \pm W\}$.*

Proof. Let $M_A = (\langle a_i, f_j\rangle)$ and $M_W = (\langle w_i, f_j\rangle)$ for an oriented orthonormal frame (f_j) of P, so that $M_A^\top M_A + M_W^\top M_W = \mathbf{1}$. The two symmetric matrices commute, so in a common eigenbasis they are $\mathrm{diag}(\cos^2\theta_i)$ and $\mathrm{diag}(\sin^2\theta_i)$, with $\theta_i \in [0, \frac{\pi}{2}]$ the principal angles between P and A. Hence $|p_A| = \prod_i \cos\theta_i$ and $|p_W| = \prod_i \sin\theta_i$. For any two indices $i \ne j$,

$$|p_A| + |p_W| \le \cos\theta_i \cos\theta_j + \sin\theta_i \sin\theta_j = \cos(\theta_i - \theta_j) \le 1.$$

Equality for every pair forces all θ_i to equal one θ, and then $\cos^4\theta + \sin^4\theta = 1 - 2\cos^2\theta\sin^2\theta = 1$ forces $\theta \in \{0, \frac{\pi}{2}\}$, that is $P = \pm A$ or $P = \pm W$. p3-14 checks the identities at 300 random planes and finds the maximum 1 of $p_A - p_W$ only at A and $-W$. □

Corollary 2.10 (The extremes of the separating overlaps; fixed)**.** *On F_3, $0 \le \mathrm{Tr}(\Pi P_L) \le 32$, with the maximum attained exactly at Π_A and Π_{-W} and the minimum exactly at Π_{-A} and Π_W. Likewise $0 \le \mathrm{Tr}(\Pi P_{\mathrm{fam}}) \le 64$, with the maximum only at Π_A and the minimum only at Π_{-A}.*

Proof. By Proposition 2.8(i) and Lemma 2.9, $|p_A - p_W| \le 1$ with $p_A - p_W = 1$ iff $P \in \{A, -W\}$ and $= -1$ iff $P \in \{-A, W\}$. Also $|p_A| \le 1$ with equality iff $P = \pm A$, since $|p_A| = \prod \cos\theta_i$. □

Proposition 2.11 (A Klein group of the projector sector; fixed)**.** *Let $\mathsf{C}(\Pi) = \mathbf{1} - \Pi$, let $\mathsf{D}(\Pi) = \frac{1}{2}(\mathbf{1} + \mathrm{vol}_+(2\Pi - \mathbf{1}))$ be the duality of Proposition 1.7, and let $\mathsf{E} = \mathsf{CD}$, so that $\mathsf{E}(\Pi) = \frac{1}{2}(\mathbf{1} + \mathrm{vol}_+) - \mathrm{vol}_+\Pi$.*

(i) $\{1, \mathsf{C}, \mathsf{D}, \mathsf{E}\}$ is a Klein four-group of K-equivariant involutions of F_3, with $\mathsf{E}(\Pi_P) = \Pi_{-P^\perp}$. It acts simply transitively on the four points: D exchanges Π_A and Π_W, C exchanges Π_A and Π_{-A}, and E exchanges Π_A and Π_{-W} (p3-16).

(ii) Each element preserves every mass matrix $\mathrm{Tr}([T, \Pi][T', \Pi])$, $T, T' \in \mathfrak{k}$, and, for connections valued in $\mathfrak{k}$ and the interior Lorentz algebra, the kinetic term and the current j_Π of every $Q \in \mathfrak{k}$ (p3-16).

(iii) E is the identity on the half $Z = +1$, the range of $\frac{1}{2}(\mathbf{1} + Z)$, which contains P_*: $\mathsf{E}(\Pi)\frac{1}{2}(\mathbf{1} + Z) = \Pi\frac{1}{2}(\mathbf{1} + Z)$, and $P_*\mathsf{E}(\Pi)P_* = P_*\Pi P_*$. On that half $\Pi_A = \Pi_{-W} = \frac{1}{2}(\mathbf{1} - \chi_W)$ and $\Pi_W = \Pi_{-A} = \frac{1}{2}(\mathbf{1} + \chi_W)$, and there $t_i^L = \Pi_A\tau_i$ and $t_i^R = \Pi_W\tau_i$. On the half $Z = -1$, $\Pi_A = \Pi_W$ and $\Pi_{-A} = \Pi_{-W}$ (p3-17).

Hence no alignment energy, kinetic term or current separates any two of the four points. No coupling that depends on Π only through $\Pi\frac{1}{2}(\mathbf{1} + Z)$ separates Π_A from Π_{-W}, or Π_{-A} from Π_W.

Proof. vol_+ is K-invariant, commutes with the interior gammas and with every Π_P, and squares to $\mathbf{1}$ (Lemma 1.2 and Proposition 1.7). C and D commute, since both $\mathsf{CD}(\Pi)$ and $\mathsf{DC}(\Pi)$ equal $\frac{1}{2}(\mathbf{1} - \mathrm{vol}_+(2\Pi - \mathbf{1}))$. (ii) For T in $\mathfrak{k}$ or in the interior Lorentz algebra, $[T, \mathsf{C}\Pi] = -[T, \Pi]$, $[T, \mathsf{D}\Pi] = \mathrm{vol}_+[T, \Pi]$ and $[T, \mathsf{E}\Pi] = -\mathrm{vol}_+[T, \Pi]$. Since vol_+ is orthogonal and commutes with $[T, \Pi]$, the traces are preserved. vol_+ is parallel for such connections, so D_μ transforms in the same way. (iii) On $Z = +1$, $\mathrm{vol}_+ = -\mathbf{1}$, so $\mathsf{E}(\Pi) = \Pi$ there. $\mathrm{vol}_A = \mathrm{vol}_+\chi_W$ equals $-\chi_W$ there and $+\chi_W$ on $Z = -1$; with $t_i^L = \frac{1}{2}(\mathbf{1} - \chi_W)\tau_i$ [1, Thm. 8.2] the remaining identities follow. □

On the half of the module that contains the matter, the vacuum Π_A is the support projector of $\mathfrak{su}(2)_L$, and Π_W that of $\mathfrak{su}(2)_R$. The residual choice is therefore, on the matter, the choice of which weak $\mathfrak{su}(2)$ acts inside the image of the vacuum. It is not a choice of spacetime chirality: the chirality halves have constant overlap, and the vector-like Q_v separates exactly as the chiral P_L does (Table 4).

Proposition 2.12 (Normalisers; fixed)**.** *(i) The fixed vectors of $\rho_+(\mathfrak{h}_{\mathrm{SM}})$ and of $\rho_+(\mathfrak{h}_{\mathrm{PS}})$ in V_+ are exactly A. A reflection of W conjugates $\mathfrak{su}(2)_L$ onto $\mathfrak{su}(2)_R$. Hence every element of K normalising $\mathfrak{h}_{\mathrm{SM}}$ fixes each of Π_A, Π_{-A}, Π_W, Π_{-W} (p3-18).*

(ii) $g = E_9E_{15} = \exp(\frac{\pi}{2}E_9E_{15})$ normalises $\mathfrak{h}_{\mathrm{PS}}$, exchanges $\mathfrak{su}(2)_L$ and $\mathfrak{su}(2)_R$, and sends $\mathrm{vol}_A \mapsto -\mathrm{vol}_A$, $\mathrm{vol}_W \mapsto -\mathrm{vol}_W$ and $X \mapsto -X$. So the normaliser of $\mathfrak{h}_{\mathrm{PS}}$ identifies Π_A with Π_{-A} and Π_W with Π_{-W}, by elements that do not preserve P_*. No element of either normaliser maps $\Pi_{\pm A}$ to $\Pi_{\pm W}$ (p3-18).

(iii) $g_{AW} = \prod_i \frac{1}{\sqrt{2}}(\mathbf{1} + a_i w_i)$, with $(a_i) = (E_2, E_3, E_8, E_9)$ and $(w_i) = (E_7, E_{13}, E_{14}, E_{15})$, is an element of $\mathrm{Spin}(8)$ with $\mathrm{Ad}(g_{AW})\mathrm{vol}_A = \mathrm{vol}_W$ and $\mathrm{Ad}(g_{AW})\mathrm{vol}_W = \mathrm{vol}_A$, which normalises neither $\mathfrak{h}_{\mathrm{SM}}$ nor $\mathfrak{h}_{\mathrm{PS}}$. Moreover $\mathrm{vol}_A\mathrm{vol}_W = \mathrm{vol}_+$ (p3-18).

Proof. (i) $\mathrm{Ad}(k)$ acts on F_3 through the rotation $R = \rho(k)$ of V_+ (Theorem 1.5). If $\mathrm{Ad}(k)$ normalises $\mathfrak{h}$, R preserves the fixed space A of $\rho_+(\mathfrak{h})$ and its complement W. For $\mathfrak{h}_{\mathrm{SM}}$, $R|_W$ normalises the derived algebra $\mathfrak{su}(2)_L$ of $\rho_+(\mathfrak{h}_{\mathrm{SM}})$. A rotation of W of determinant -1 is a rotation times a reflection, and it maps $\mathfrak{su}(2)_L$ onto $\mathfrak{su}(2)_R$; so $\det R|_W = 1$, hence $\det R|_A = 1$, and R fixes the oriented planes A and W. (ii) $\mathrm{Ad}(g)$ is the rotation by π in the plane (E_9, E_{15}), of determinant -1 on A and on W; it preserves W and V_-, hence $\mathfrak{h}_{\mathrm{PS}}$. Γ commutes with g and $\mathrm{Ad}(g)\chi_W = -\chi_W$. The last statement follows from (i), which applies to $\mathfrak{h}_{\mathrm{PS}}$ as well. (iii) $\mathrm{Ad}(\exp\frac{\pi}{4}a_iw_i)$ rotates the plane (a_i, w_i) by $\frac{\pi}{2}$; the signs and the failure to normalise are computed in p3-18. □

With $\mathfrak{h}_{\mathrm{SM}}$ gauged the four points are therefore distinct points of the gauged theory, although the projector sector cannot tell them apart. With $\mathfrak{h}_{\mathrm{PS}}$ gauged, Π_A and Π_{-A} are related by a left–right exchange of the Pati–Salam algebra, which the matter block breaks.

2.5 Robustness

Lemma 2.13 (Invariant couplings keep fixed critical points; inherited, with proof)**.** *Let $H \subset K$ be connected with Lie algebra $\mathfrak{h}$, let f be a smooth H-invariant function on F_3, and let $\Pi_0 \in F_3$ be fixed by H. Then df_{Π_0} is an H-invariant covector. If $T_{\Pi_0}F_3$ contains no nonzero $\mathfrak{h}$-invariant vector, Π_0 is a critical point of f. In particular Π_0 stays critical for $V_0 + tf$ for every t whenever V_0 is H-invariant too, and it stays a nondegenerate local minimum as long as the Hessian of $V_0 + tf$, an H-invariant symmetric form, stays positive definite.*

Proof. For $h \in H$, $f \circ h = f$ and $h\Pi_0 = \Pi_0$ give $df_{\Pi_0} \circ dh_{\Pi_0} = df_{\Pi_0}$. The metric $\mathrm{Tr}(d\Pi\, d\Pi)$ is K-invariant (Lemma 1.12), so the gradient is an H-invariant tangent vector, and for a connected group the invariant vectors are those killed by $\mathrm{ad}\,\mathfrak{h}$ [4, 5]. The Hessian at a critical point is invariant for the same reason. □

Proposition 2.14 (Invariant tangent vectors; fixed)**.** *The dimensions of the $\mathfrak{h}$-invariant subspaces of the tangent spaces at Π_A and Π_W, and so at Π_{-A} and Π_{-W}, whose tangent spaces are the same up to sign, are: $\mathfrak{h}_{\mathrm{SM}}$, $\mathfrak{h}_{\mathrm{PS}}$, $\mathfrak{su}(2)_L$ and $\mathfrak{u}(1)_Y \oplus \mathfrak{su}(3)_C$: 0; $\mathfrak{u}(1)_{\mathrm{em}} \oplus \mathfrak{su}(3)_C$: 8, the neutral components $A \otimes W_0$; $\mathfrak{u}(1)_A$: 8; $\mathfrak{u}(1)_A \oplus \mathfrak{u}(1)_{\mathrm{em}}$: 4 (p3-19).*

Corollary 2.15 (Matter couplings that respect $\mathfrak{h}$; fixed, with an inherited principle)**.** *Let the vacuum energy on F_3 be the sum of an alignment energy and of couplings to matter that respect $\mathfrak{h} \in \{\mathfrak{h}_{\mathrm{SM}}, \mathfrak{h}_{\mathrm{PS}}\}$: functions of Π and of operators fixed by $\mathfrak{h}$, such as the overlaps*

of Proposition 2.8, Yukawa-type kernels commuting with $\mathfrak{h}$, or the fermion loop of a Dirac operator with $\mathfrak{h}$-invariant couplings [1, Section 21]. Such an energy is H-invariant, since $f(\mathrm{Ad}(h)\Pi; M) = f(\Pi; \mathrm{Ad}(h^{-1})M)$, and $\Pi_{\pm A}$, $\Pi_{\pm W}$ remain critical points of it. All linear overlaps of Table 4, including the one with P_{fam}, which does not commute with $\mathfrak{su}(3)_C$, are $\mathfrak{h}_{\mathrm{SM}}$-invariant functions on F_3, and their first variations vanish at the four points along all of $\mathfrak{k}$ (p3-20). With only $\mathfrak{u}(1)_{\mathrm{em}} \oplus \mathfrak{su}(3)_C$ gauged the eight neutral directions are invariant, and an invariant coupling can move the vacuum along the $\mathfrak{u}(1)_{\mathrm{em}}$-fixed component $\widetilde{\mathrm{Gr}}_4(A \oplus W_0)$ of Theorem 1.16.

Proof. Lemma 2.13 with Proposition 2.14. The invariance of the overlaps is p3-20; for P_{fam} it also follows from $\mathrm{Tr}(\Pi_P P_{\mathrm{fam}}) = \frac{1}{4}\mathrm{Tr}(\Pi_P \Pi_A)$ and $[\mathfrak{h}_{\mathrm{SM}}, \Pi_A] = 0$. □

What invariant couplings can do is change the Hessian. Write $\Pi(x) = \mathrm{Ad}(e^X)\Pi_0$ with $X = \sum x_{aw} \frac{1}{2} aw$, as in Theorem 1.10, and

$$c_{\mathrm{SM}} = 6g_2^2 + \tfrac{6}{5}g_Y^2, \qquad c_{\mathrm{PS}} = 12g^2.$$

Proposition 2.16 (Hessians at the four points; fixed)**.** *At each of the four points $V_{\mathfrak{h}_{\mathrm{SM}}}(\Pi(x)) = 128\, c_{\mathrm{SM}}|x|^2 + O(x^3)$ and $V_{\mathfrak{h}_{\mathrm{PS}}}(\Pi(x)) = 128\, c_{\mathrm{PS}}|x|^2 + O(x^3)$. The overlaps expand as*

$$\mathrm{Tr}(\Pi(x)P_L) = \begin{cases} 32 - 8|x|^2 + O(x^3) & \text{at } \Pi_A,\ \Pi_{-W}, \\ 8|x|^2 + O(x^3) & \text{at } \Pi_{-A},\ \Pi_W, \end{cases} \qquad \mathrm{Tr}(\Pi(x)P_{\mathrm{fam}}) = \begin{cases} 64 - 16|x|^2 + \cdots & \text{at } \Pi_A, \\ 16|x|^2 + \cdots & \text{at } \Pi_{-A}, \\ 32 + O(x^4) & \text{at } \Pi_{\pm W} \end{cases}$$

(p3-21). For $V = \kappa V_{\mathfrak{h}_{\mathrm{SM}}} - \lambda\,\mathrm{Tr}(\Pi P_L) - \mu\,\mathrm{Tr}(\Pi P_{\mathrm{fam}})$ every Hessian at the four points is a multiple of the identity. Π_A and Π_{-W} are nondegenerate local minima for all $\kappa \geq 0$, $\lambda, \mu \geq 0$ not all zero. Π_W is a local minimum iff $\lambda < 16\kappa c_{\mathrm{SM}}$, and Π_{-A} iff $\lambda + 2\mu < 16\kappa c_{\mathrm{SM}}$. Beyond these values all sixteen directions become unstable at once.

Proof. Since $[T_a, \Pi_0] = 0$, $\|[T_a, \Pi(x)]\|^2 = \|[[T_a, X], \Pi_0]\|^2 + O(x^3)$. $[T_a, X]$ corresponds to $x\,\rho_+(T_a)^\top$ on $A \otimes W$, so the sum is $128\ \mathrm{tr}(x\,C\,x^\top)$ with $C = \sum_a g_a^2 \rho_+(T_a)\rho_+(T_a)^\top$ on W. For $\mathfrak{h}_{\mathrm{SM}}$ the Casimir of $\mathfrak{su}(2)_L$ and $\rho_+(t_Y)^2 = -\frac{1}{4}$ give $C = (6g_2^2 + \frac{6}{5}g_Y^2)\mathbf{1}_W$, and $\mathfrak{su}(3)_C$ gives nothing; for $\mathfrak{h}_{\mathrm{PS}}$ each of the four directions of W lies in three unit blades, so $C = 12g^2\mathbf{1}_W$. The overlaps follow from Proposition 2.8(i) with $p_A(\Pi(x)) = 1 - \frac{1}{2}|x|^2 + O(x^4)$ near A, and $p_W = \det(O(x)) = O(x^4)$ there; near $\pm W$ the roles are exchanged. p3-21 checks the second derivatives $\mathrm{Tr}([X, [X, \Pi_0]]P)$ exactly and by finite differences. The stability conditions compare $128\kappa c_{\mathrm{SM}}$ with 8λ, and with $8\lambda + 16\mu$ at Π_{-A}. □

The sign of λ is a choice. For $\lambda < 0$ the roles of $\{\Pi_A, \Pi_{-W}\}$ and $\{\Pi_{-A}, \Pi_W\}$ are exchanged. The Hessians found here are the multiples of the identity in the sixteen-dimensional space of $\mathfrak{h}_{\mathrm{SM}}$-invariant forms of Corollary 1.18; a coupling with a component of the form $N \otimes \omega_Y$ would split the doublet directions.

2.6 The complete selection

The alignment result used is Proposition 1.15 and Theorem 1.16: for positive couplings $V_{\mathfrak{h}_{\mathrm{SM}}} \geq 0$ on F_3, with zero set exactly $\{\Pi_{\pm A}, \Pi_{\pm W}\}$.

Theorem 2.17 (Selection on F_3)**.** *Assume:*

(H1) the vacuum projector is a point of F_3 (constructed: Section 1.6);

(H2) $\mathfrak{h}_{\mathrm{SM}}$ is gauged, embedded as in [1, Section 8], with couplings $g_3, g_2, g_Y > 0$ (constructed);

(H3) the sign hypothesis: the gauge fluctuations contribute $\kappa V_{\mathfrak{h}_{\rm SM}}$ to the vacuum energy with $\kappa \geq 0$ (inherited mechanism of vacuum alignment [6, 7]; its sign is proved for vector-like theories [14] and assumed here);

(H4) the vacuum couples to the block by $-\lambda \operatorname{Tr}(\Pi P_L)$ with $\lambda > 0$ (constructed, sign chosen);

(H5) optionally, it couples to the family block by $-\mu \operatorname{Tr}(\Pi P_{\rm fam})$ with $\mu > 0$ (constructed).

Let $V = \kappa V_{\mathfrak{h}_{\rm SM}} - \lambda \operatorname{Tr}(\Pi P_L) - \mu \operatorname{Tr}(\Pi P_{\rm fam})$, with $\mu = 0$ when (H5) is not assumed.

(a) Under (H1)–(H3) the minimum set is $\{\Pi_{\pm A}, \Pi_{\pm W}\}$. Overlaps with P_*, $P_{\rm lep}$ and $\frac{1}{2}(\mathbf{1} \pm \Gamma)$, linear or quadratic, take equal values at these points and do not separate them.

(b) Under (H1)–(H4) the global minimum value is -32λ, attained exactly at Π_A and Π_{-W}, for every $\kappa \geq 0$. These two points coincide on the half $Z = +1$ that contains P_*, and they are exchanged by E, which preserves every term of V except the one of (H5). Π_W and Π_{-A} remain critical, and are local minima iff $\lambda < 16\kappa c_{\rm SM}$.

(c) Under (H1)–(H5) the global minimum is Π_A alone, with value $-32\lambda - 64\mu$.

(d) At the selected vacuum all of $\mathfrak{h}_{\rm SM}$ is unbroken, the massless vector of Theorem 2.6(a) is the photon of $\mathfrak{u}(1)_{\rm em}$, and the field along Q_A is a second abelian field.

Proof. (a) is Proposition 1.15 and Theorem 1.16 and Table 4. (b) $V \geq \kappa \cdot 0 - 32\lambda$ by Corollary 2.10, with equality iff $V_{\mathfrak{h}_{\rm SM}} = 0$ and $\operatorname{Tr}(\Pi P_L) = 32$, that is iff $\Pi \in \{\Pi_A, \Pi_{-W}\}$, where both conditions hold. The coincidence and the role of E are Proposition 2.11; the overlap with P_L depends on Π only through $\Pi\frac{1}{2}(\mathbf{1} + Z)$. Stability is Proposition 2.16. (c) With $\mu > 0$ equality requires in addition $\operatorname{Tr}(\Pi P_{\rm fam}) = 64$, which holds only at Π_A (Corollary 2.10). (d) is Theorem 1.9 and Theorem 2.6. p3-22 minimises V from random starts for several couplings, including $\lambda > 16\kappa c_{\rm SM}$ and $\kappa = 0$, and finds exactly these minima. □

Two features of the theorem could have come out otherwise. The overlap with P_* itself, which the matter block suggests first, is constant on F_3 and selects nothing, and the selection is made by the weak-chirality half P_L. Once that overlap is present the alignment energy is not needed for the selection; it only stabilises the other two critical points. The pair $\{\Pi_A, \Pi_{-W}\}$ is one projector on the half of the module that carries the matter, so for physics on the block the selected vacuum is unique (interpretive). A unique point of F_3 needs a coupling that sees the other half. The overlap with $P_{\rm fam}$ is such a coupling, but $P_{\rm fam}$ carries no colour, and adopting it amounts to choosing Π_A, since $\operatorname{Tr}(\Pi P_{\rm fam}) = \frac{1}{4}\operatorname{Tr}(\Pi\Pi_A)$ on F_3 (constructed). The rank-256 projector Π_A is the projector P_1 of the projector family [1, Thm. 7.5], on which colour and a chiral doublet first coexist [1, Thms. 7.7, 7.9].

Proposition 2.18 (Restated data of the projectors; restated)**.** *$X_1 = J^{(1)}J^{(2)} = \mathrm{vol}_A$, so $P_1 = \Pi_A$, with $\dim \mathfrak{g}_{\rm phys}(P_1) = 51$, $\dim \mathfrak{k}(P_1) = 27$ and a three-dimensional annihilator, as in [1, Thm. 7.5], and $P_{\rm fam} \leq \Pi_A$. The projector $P_{\rm lep}$ of Definition 2.1 has rank 64, commutes with the interior gammas and with J, and with P_*, $P_{\rm fam}$ and Q_v reproduces the alignment energies 3, 5, 9, 5 and stabilisers 37, 23, 9, 25 of [1, Prop. 21.6] (p3-15).*

Proof. $X_1 = e_2^{(1)}e_3^{(1)}e_2^{(2)}e_3^{(2)} = E_2E_3E_8E_9$. The other statements are the computation p3-15. □

2.7 The answer

(a) *Which $\mathfrak{u}(1)$ is the photon.* At the third rung the photon of the aligned vacua is the $\mathfrak{u}(1)_{\rm em}$ gauge field of the internal connection, generated by $B_{\rm em} = -\frac{1}{2}E_7E_{13} + \frac{1}{6}H_C$. It is massless at $\Pi_{\pm A}$ and $\Pi_{\pm W}$, its projector current is carried by four complex scalars of the electron's charge magnitude, and on the block it couples with the Standard-Model charges without anomaly (fixed, given the adopted embedding, which is constructed).

(b) *The construction's charge.* $Q = \frac{1}{2}J_1$ becomes $Q_A = \frac{1}{2}E_2E_3$, whose charge operator is $\frac{1}{2}U$. On the block it is half the fermion number, the same on every state, and anomalous with $\mathfrak{su}(2)_L$ (fixed; anomaly restated). It is not electromagnetism (obstructed). The normal $SO(2)$ connection that carries it is a separate abelian field (fixed), and its coupling is not decided (open).

(c) *The counts and the dichotomy.* Of the three counts against the normal $U(1)$, the spinless count is unchanged, the adapted-generator count is removed for both charges, and the sign count changes from forced to adopted; a fourth count, the charge on the block, decides. Both alternatives of the dichotomy of [1, Section 25] hold at $n = 3$ (fixed; the Kaluza–Klein reading of the sign count is interpretive).

(d) *Which vacuum.* Alignment by $\mathfrak{h}_{\mathrm{SM}}$ leaves $\Pi_{\pm A}, \Pi_{\pm W}$ (fixed, Theorem 1.16). A Klein group of symmetries of the projector sector permutes them simply transitively; no alignment energy, kinetic term or current separates them (fixed). The weak-chirality overlap $-\lambda \operatorname{Tr}(\Pi P_L)$, $\lambda > 0$, selects $\{\Pi_A, \Pi_{-W}\}$, which is one projector on the matter half (fixed under constructed hypotheses). A single point requires a coupling off that half, such as $-\mu \operatorname{Tr}(\Pi P_{\mathrm{fam}})$, which selects Π_A (constructed).

(e) *Robustness.* $\mathfrak{h}_{\mathrm{SM}}$- or $\mathfrak{h}_{\mathrm{PS}}$-invariant couplings, including matter couplings, keep the four points critical, because their tangent spaces carry no invariant vector (fixed; principle inherited). They can only change the Hessians, which for the couplings considered are multiples of the identity; Π_W and Π_{-A} lose stability for $\lambda > 16\kappa c_{\mathrm{SM}}$ and $\lambda + 2\mu > 16\kappa c_{\mathrm{SM}}$ (fixed).

(f) *Open.* The dynamical origin and sign of the overlap couplings, the scales f, g, κ, λ, μ, the coupling of the second abelian field, whether the half $Z = -1$ is physical, and the origin of the Maxwell term of $\mathfrak{u}(1)_{\mathrm{em}}$ are open.

References

[1] D. Bencheikh, *The electromagnetic sphere: structure, matter, dynamics and causal structure of the split-signature Clifford algebra* $\mathrm{Cl}(3,3)$ *and its tower*, 17 October 2026 (`ES_electromagnetic_sphere_2026-10-17.pdf`, sources in `tex/`).

[2] *The undivided energy and its differentiation:* $E = mc^2$ *in the split-signature arena*, investigation note to [1], 19 October 2026 (`investigations/rest_energy/main.pdf`).

[3] S. Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, American Mathematical Society, 2001.

[4] L. Michel, *Points critiques des fonctions invariantes sur une G-variété*, C. R. Acad. Sci. Paris Sér. A **272** (1971) 433.

[5] R. S. Palais, *The principle of symmetric criticality*, Commun. Math. Phys. **69** (1979) 19.

[6] M. E. Peskin, *The alignment of the vacuum in theories of technicolor*, Nucl. Phys. B **175** (1980) 197.

[7] J. Preskill, *Subgroup alignment in hypercolor theories*, Nucl. Phys. B **177** (1981) 21.

[8] C. Bouchiat, J. Iliopoulos and Ph. Meyer, *An anomaly-free version of Weinberg's model*, Phys. Lett. B **38** (1972) 519.

statement	class	checks
Definition 2.1, Proposition 2.18: P_{lep}, $P_1 = \Pi_A$, the table of ES17 Prop. 21.6	restated (ES17 Thm. 7.5, Prop. 21.6)	p3-15
Proposition 2.2: $\hat{Q}_A = \frac{1}{2}U$, $q_A P_* = J P_*$, charge table, Weyl block	fixed; $q_A P_* = J P_*$ and the $\hat{Y}$, $\hat{Q}_{\mathrm{em}}$ columns restated	p3-1 to p3-3
Corollary 2.3: Q_A is $\frac{1}{2}F$ on the block, not the electric charge, anomalous; $\mathfrak{u}(1)_{\mathrm{em}}$ anomaly free	fixed; obstructed; anomaly restated; anomaly conditions inherited	p3-4, p3-5
Table 3: the four counts and the dichotomy	restated (i); fixed (ii), (iv); constructed and interpretive (iii)	p3-4, p3-5
Proposition 2.4: joint fixed points of Q_A and B_{em}	fixed	p3-6
Proposition 2.5: torus weights of the fluctuations	fixed	p3-7
Theorem 2.6: the photon is $\mathfrak{u}(1)_{\mathrm{em}}$; common charge unit; Q_A field separate; m_W^2/m_Z^2	fixed; gauged generator constructed; (b) obstructed	p3-4, p3-8, p3-9
Lemma 2.7, Proposition 2.8: overlaps as Plücker functions, values, commutant of P_*	fixed	p3-10 to p3-13
Lemma 2.9, Corollary 2.10: comass inequality, extremes of the separating overlaps	fixed	p3-12, p3-14
Proposition 2.11: Klein group, E trivial on the matter half	fixed	p3-16, p3-17
Proposition 2.12: normalisers of $\mathfrak{h}_{\mathrm{SM}}$, $\mathfrak{h}_{\mathrm{PS}}$; g_{AW}	fixed	p3-18
Lemma 2.13: invariant couplings keep fixed critical points	inherited [4, 5], proof given	—
Proposition 2.14, Corollary 2.15: invariant tangent vectors, matter couplings	fixed; invariance of induced energies inherited	p3-19, p3-20
Proposition 2.16: Hessians and stability conditions	fixed	p3-21
Theorem 2.17: selection of $\{\Pi_A, \Pi_{-W}\}$, and of Π_A with (H5)	fixed under (H1), (H2), (H4), (H5) constructed and (H3) inherited	p3-22
unique vacuum for physics on the block; Kaluza–Klein reading of the sign count	interpretive	—
origin and sign of the overlap couplings; scales; the second abelian field	open	—

Table 5: Claims of Section 2 and their classes. The checks are the `PASS` lines printed by `verify/p3_identification.py`.

[9] H. Georgi and S. L. Glashow, *Gauge theories without anomalies*, Phys. Rev. D **6** (1972) 429.

[10] G. 't Hooft, *Symmetry breaking through Bell–Jackiw anomalies*, Phys. Rev. Lett. **37** (1976) 8.

[11] C. M. Hull, *Timelike T-duality, de Sitter space, large N gauge theories and topological field theory*, JHEP **07** (1998) 021.

[12] S. L. Glashow, *Partial-symmetries of weak interactions*, Nucl. Phys. **22** (1961) 579.

[13] S. Weinberg, *A model of leptons*, Phys. Rev. Lett. **19** (1967) 1264.

[14] E. Witten, *Some inequalities among hadron masses*, Phys. Rev. Lett. **51** (1983) 2351.